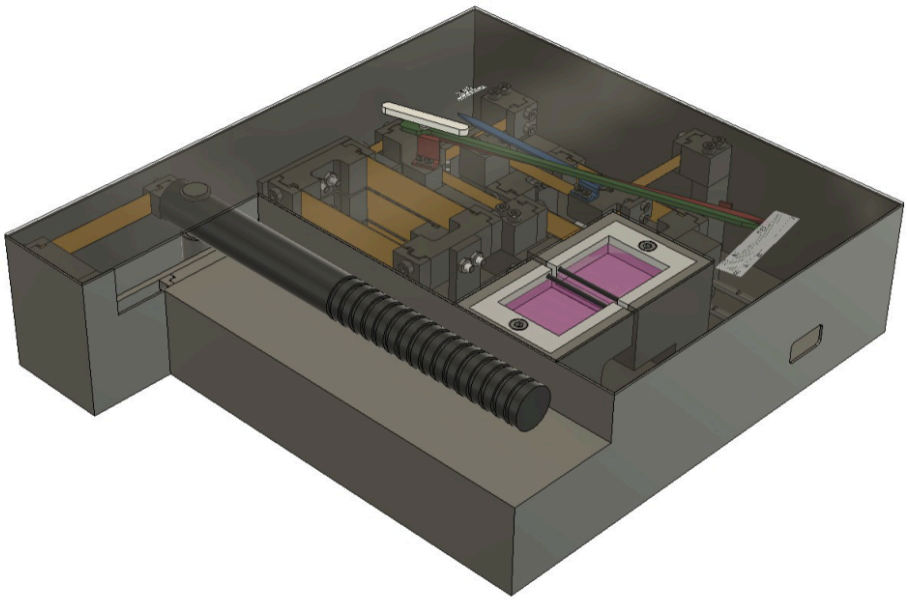
StitchMeter

Mechanism Design — Group 32

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1 Introduction

The goal of the StitchMeter project is to design a device that measures the tightening force of surgical stitches and then tests their strength up to failure. Proper knot tension is important for good wound healing, so it is an essential skill for surgeons to acquire. This device would let surgeons train easily, particularly in disaster areas where conditions can be harsh. This is why the mechanism must be purely mechanical and robust, so that it remains functional for as long as possible. Flexure guides are highly resistant to wear and require very little maintenance, which makes them particularly well suited to such a mechanism.

2 General Design

The system consists mainly of two moving blocks: block A[1], which moves mainly during phase I (and also during phase II if the wound relaxes), and block B, which moves during phase II.

2.1 Guiding principle for the wound edges

Only one of the two wound edges is mobile, while the other is fixed to the frame. The mobile edge is attached to block A, which is itself guided by a parallel-blade stage. Since the parallel-blade stage only allows a translation, the wound edges always remain parallel. There is, however, a parasitic motion λ, so the alignment of the ends of the wound is not guaranteed.

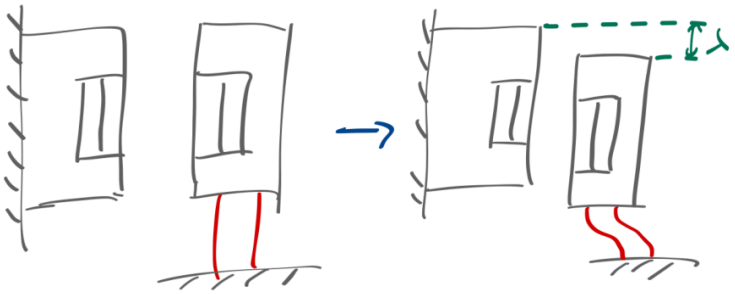


Figure 1: Parasitic motion while the wound edges are brought together

[1] Blocks A and B correspond to the coloured blocks shown in the mechanism diagrams in sections 2.9 and 2.10.

2.2 Force measurement principle during phase I

During phase I, the knot brings the two wound edges together (which amounts to bringing block A closer to the frame). Since block B is held against a stop, only block A is mobile during this phase. Knowing the equivalent stiffness of the system, the force exerted by the knot can therefore be deduced from the displacement of block A relative to the frame.

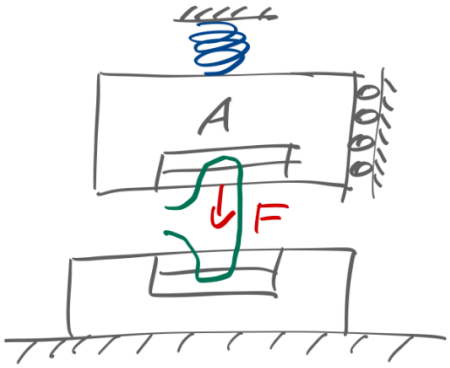


Figure 2: Force measurement during phase I

2.3 Display amplification principle during phase I

The 2 [mm] displacement is too small for the required resolution, so it must be amplified. Through a link, block A is connected to a needle pivoting on the frame, which turns the translation of the wound edges into a rotation of the needle. The attachment point of the link on the needle and the length of the needle are then sized to obtain the desired amplification.

2.4 Force measurement principle during phase II

During phase II, block A and the frame are connected by the thread. The thread blocks the only degree of freedom of block A, which makes block A rigidly linked to the frame. When a force is applied to block A (in the direction of the thread), this force therefore "flows" through the thread to the frame. Block B, which is mobile and separated from block A by several blade stages, makes it possible to apply this force in a controlled way: the closer block B moves to block A, the more the blade stages are deformed, and so the larger the force that "pushes" block A and passes through the thread. Knowing the equivalent stiffness of the blade stages between blocks A and B, measuring the relative distance between the two blocks is enough to know the force passing through the thread.

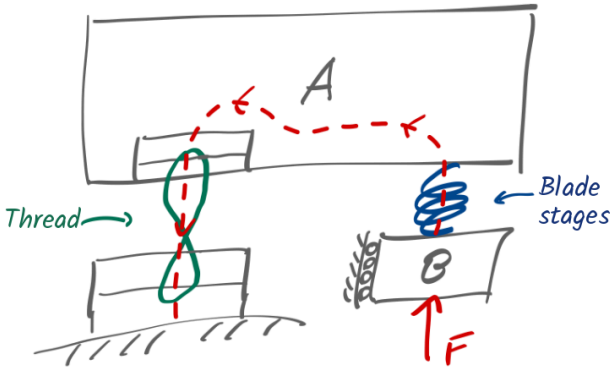


Figure 3: "Path" followed by the force

2.5 Display amplification principle during phase II

The same principle as in phase I is used, this time to measure the distance between block A and block B.

2.6 Drag needle principle during phase II

The drag needle is simply mounted on the frame as a plain pivot (shaft in a hole), without blade stages, and is pushed by the main needle. Since the main needle is driven by the distance between blocks A and B, when the thread breaks or suddenly relaxes, the main needle swings back (blocks A and B move apart under the action of the pushing blade stages) and leaves the drag needle in place, showing the maximum force reached.

2.7 Motion transmission from the actuation lever, with force amplification

The force is applied to the system through a lever. To test the thickest thread up to failure, a force of about 100 [N] must be applied to it. So that the actuation force does not exceed 10 [N], the lever is sized to amplify the input force by a factor of 12.

2.8 Stiffness adjustment for the three thread types

During phase I, the maximum force to be measured depends on the thread being tested. To change the stiffness of the system, the blade stage guiding block A can be preloaded more or less, using a helical spring and a screw that controls its extension. The equivalent stiffness of the system can thus be changed in a controlled way, so that for the same 2 [mm] stroke, the maximum measured force matches the thread being tested.

2.9 Mechanism diagram with ideal guides and ideal springs

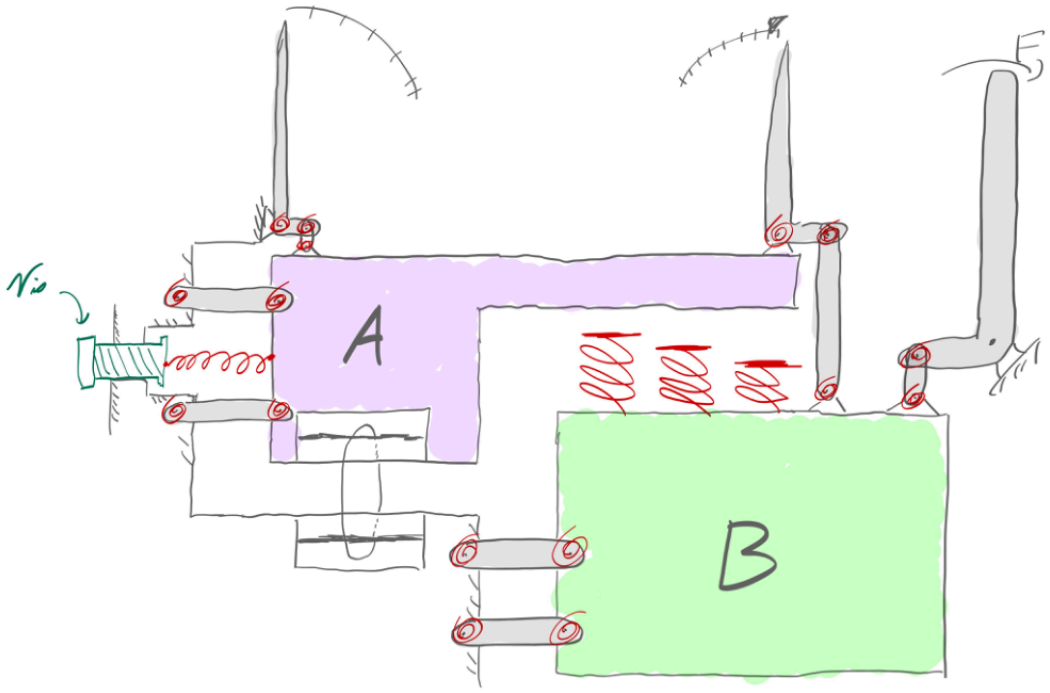


Figure 4: Mechanism diagram with ideal guides

2.10 Mechanism diagram with flexure guides

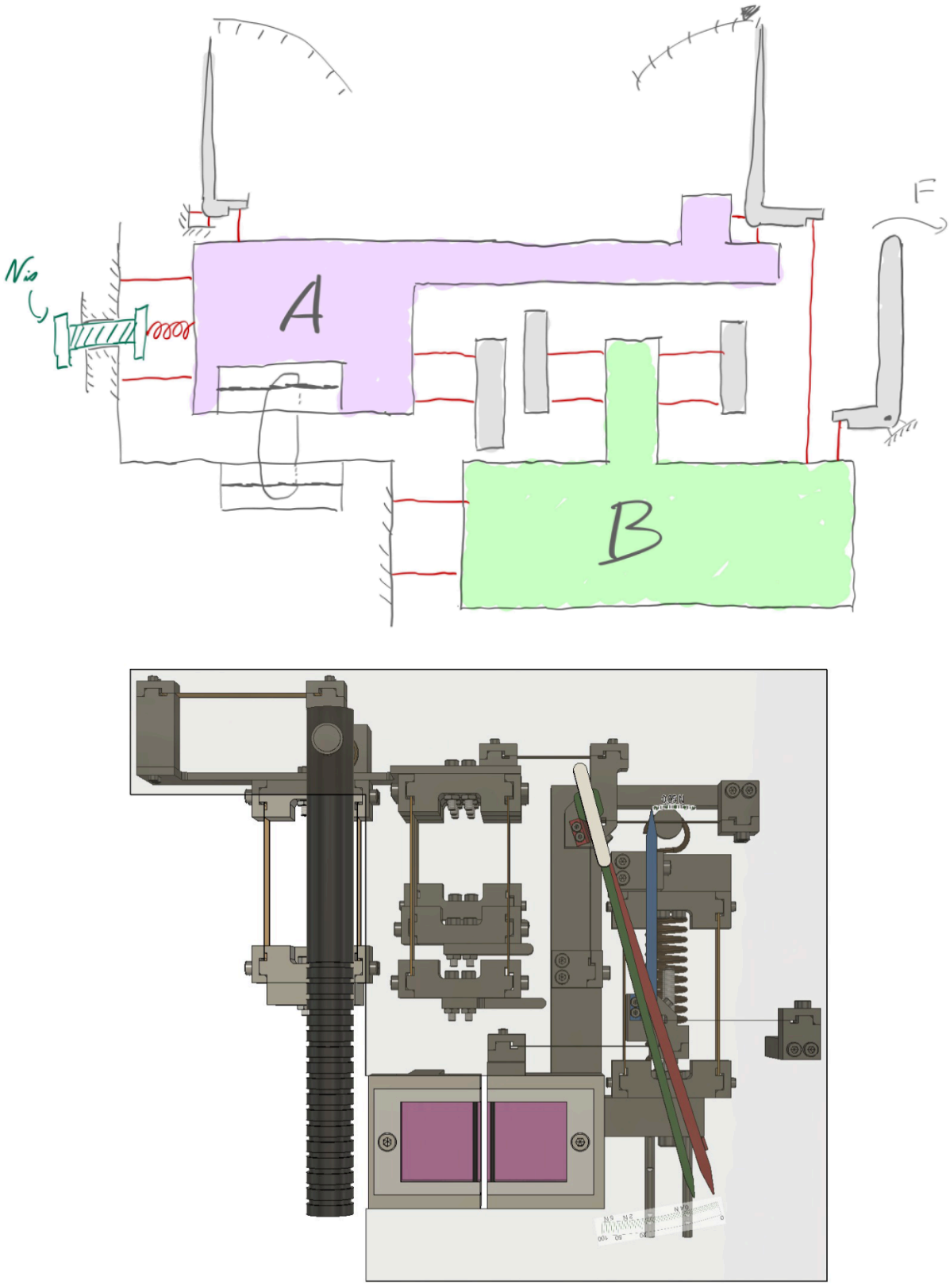


Figure 5: Diagram and CAD of the mechanism with flexure guides

3 Detailed Sizing

3.1 Stiffness of the measuring elements

In the rest of this document, specific flexure elements are referred to by name (pivot I, link I, etc.). The reference of each flexure element is given in the appendix at the end of the document (Figure 11).

3.1.1 Phase I

During phase I, the system must be sized so that forces of up to one tenth of the nominal strength of the thread $F_{r,\text{nominal}}$ can be measured, with a maximum closing of the wound of 2 [mm]. Using the relation

$$F = kx$$

where F, k and x are respectively the measured force [N], the equivalent stiffness of the system [N/mm] and the displacement [mm], the required equivalent stiffness is easily calculated for the three threads:

Thread	$F_{r,\text{nominal}}/10$ **[N]**	x_{max} **[mm]**	k_{eq} **[N/mm]**
EP3	1	2	0.5
EP6	4	2	2
EP9	8	2	4

The flexure guides involved in the equivalent stiffness are pivots I and II, blade stage I, and links I and II. Since the stiffness of the blade stage is variable, it will be calculated separately later. The links are approximated as single-blade RCC pivots (see figure below).

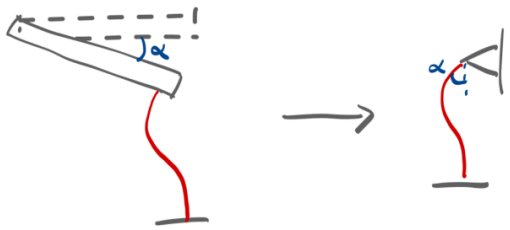


Figure 6: Approximation of the link as a single-blade RCC pivot

By conservation of energy:

$$\frac{1}{2}k'_{eq}x^2 = \frac{1}{2}k_{\text{pivot I}}\alpha_1^2 + \frac{1}{2}k_{\text{pivot II}}\alpha_2^2 + \frac{1}{2}k_{\text{link I}}\alpha_3^2 + \frac{1}{2}k_{\text{link II}}\alpha_4^2$$

Expressing $\alpha_{1,2,3,4}$ as functions of x and using the small-angle approximation $\sin(\alpha) \simeq \alpha$, the equation above becomes:

$$\frac{1}{2}k'_{eq}x^2 = \frac{1}{2}k_{\text{pivot I}}\left(\frac{x}{r_1}\right)^2 + \frac{1}{2}k_{\text{pivot II}}\left(\frac{x}{r_2}\right)^2 + \frac{1}{2}k_{\text{link I}}\left(\frac{x}{r_3}\right)^2 + \frac{1}{2}k_{\text{link II}}\left(\frac{x}{r_4}\right)^2$$

which, once rearranged and simplified, gives:

$$k'_{eq} = \frac{k_{\text{pivot I}}}{r_1^2} + \frac{k_{\text{pivot II}}}{r_2^2} + \frac{k_{\text{link I}}}{r_3^2} + \frac{k_{\text{link II}}}{r_4^2}$$

Using the stiffness formula for a two-blade RCC pivot[2] (with a coefficient of 0.5 for the links, which have only one blade):

$$k_\theta = \frac{8EI\left(l^2 + 3pl + 3p^2\right)}{l^3} \qquad \text{where} \qquad I = \frac{bh^3}{12}$$

The equivalent stiffness of the pivots and links k'_{eq} can be calculated from the values in the table below:

	E [MPa]	**l [mm]**	**b [mm]**	**h [mm]**	**p [mm]**	**r [mm]**	**k [N/mm]**
Pivot I	193000	40	6	0.1	10	10	0.374
Pivot II	193000	40	6	0.1	10	25	0.060
Link I	72000	40	5	0.1	0	10	0.030
Link II	193000	40	5	0.1	0	25	0.014
Equivalent stiffness k'_{eq}							0.478

From there, the required stiffnesses of blade stage I are easily calculated with the relation:

$$k_{eq} = k_{\text{stage I}} + k'_{eq}$$

which gives the following values:

Thread	**EP3**	**EP6**	**EP9**
Stage stiffness $k_{\text{stage},i}$ [N/mm]	0.022	1.522	3.522

These stiffness values can be obtained with a single blade stage by varying its preload. The stiffness k_{stage} of a blade stage preloaded by a force N is given by[3]:

$$k_{\text{stage I}} = k_0 - \frac{k_0}{N_0}N \qquad \text{where} \qquad k_0 = \frac{24EI}{l^3}, \quad N_0 = \frac{2\pi^2 EI}{l^2}, \quad I = \frac{bh^3}{12}$$

The preload needed to reach the different target stiffnesses can thus be calculated from the dimensions of blade stage I, which are set as follows:

E [MPa]	l [mm]	b [mm]	h [mm]	k_0 [N/mm]	N_0 [N]
193000	50	10	0.5	3.86	158.7

The preload N must then be:

Thread	EP3	EP6	EP9
Preload N [N]	157.8	96.1	13.9

This force is applied by a helical spring whose extension is adjusted by the user depending on the thread being tested.

[2,3] Formulas from the book "Conception des guidages flexibles" [1].

3.1.2 Sizing of the preload spring

The preload N is applied to blade stage I by a helical tension spring whose extension is controlled by an adjustable screw. The spring must be able to deliver a maximum force $N_{max} = 157.8$ [N] for the EP3 thread while remaining in its elastic range.

The spring dimensions were chosen to obtain a stiffness k_{spring} such that the extensions δ remain in a range usable by the operator: not so small that precision is lost, not so large that the device becomes bulky. We therefore chose a spring of stiffness $k_{\text{spring}} = 4.55$ [N/mm], free length $L_0 = 74.20$ [mm] and maximum allowable length $L_n = 111.20$ [mm], which gives an allowable stroke of:

$$L_n - L_0 = 111.20 - 74.20 = 37 \text{ [mm]}$$

The spring extension needed for each thread is:

$$\delta = \frac{N_{\text{thread}}}{k_{\text{spring}}}$$

The resulting values are given in the following table:

Thread	EP3	EP6	EP9
Preload N [N]	157.8	96.1	13.9
Extension δ [mm]	34.7	21.1	3.1

All three extensions are below the 37 [mm] allowable stroke of the spring, so the chosen spring is compatible with all three threads. These three settings are marked directly on the frame to make the device easier to use.

3.1.3 Sizing of the preload screw

The preload screw controls the extension of the helical spring, and therefore the preload applied to blade stage I. It is subjected to a maximum axial tensile load equal to the force applied for the EP3 thread, i.e. $N = 157.8$ [N].

Choice and strength. The selected screw is an M4 $\times$ 0.7 in A2-70 stainless steel. The M4 diameter was chosen for the adjustment resolution it offers: with a pitch $p = 0.7$ [mm], the change in preload per turn is $\Delta N = 3.185$ [N/turn], which makes it easy to switch from one thread setting to another in about 20 turns.

The standard tensile stress area of an M4 $\times$ 0.7 thread is $A_s = 8.78$ [mm²]. The maximum tensile stress is therefore:

$$\sigma = \frac{N}{A_s} = \frac{157.8}{8.78} = 18.0 \ [\text{MPa}]$$

A2-70 stainless steel has a proof stress $R_{p0.2} = 405$ [MPa], which gives a safety factor of:

$$s = \frac{R_{p0.2}}{\sigma} = \frac{405}{18.0} \approx 22.5$$

The screw is therefore largely oversized with respect to mechanical strength.

Self-locking. The screw must be self-locking so that the preload stays stable during use. The helix angle λ is the inclination of the thread spiral relative to a plane perpendicular to the screw axis. It is calculated from the pitch p and the pitch diameter d_2:

$$d_2 = d - 0.6495 \times p = 4 - 0.6495 \times 0.7 = 3.545 \ [\text{mm}]$$

$$\lambda = \arctan\left(\frac{p}{\pi d_2}\right) = \arctan\left(\frac{0.7}{\pi \times 3.545}\right) = 3.6^\circ$$

For a steel-on-steel friction coefficient $\mu = 0.15$, the friction angle is:

$$\varphi = \arctan(\mu) = \arctan(0.15) = 8.5^\circ$$

The self-locking condition $\lambda \leq \varphi$ is satisfied ($3.6^\circ \leq 8.5^\circ$), so no additional thread-locking device is needed.

Force applied to the screw. The actual efficiency of the thread is calculated from the angles λ and φ:

$$\eta = \frac{\tan(\lambda)}{\tan(\lambda + \varphi)} = \frac{\tan(3.6^\circ)}{\tan(3.6^\circ + 8.5^\circ)} = \frac{0.063}{0.215} = 0.29$$

The torque needed to turn the screw is:

$$C = \frac{N \cdot p}{2\pi \cdot \eta}$$

The tangential force to apply on the knurled knob of diameter $d_{\text{knob}} = 30$ [mm] is:

$$F_{\text{knob}} = \frac{2C}{d_{\text{knob}}}$$

These values are summarised for the three threads in the following table:

Thread	**EP3**	**EP6**	**EP9**
Preload N [N]	157.8	96.1	13.9
Torque C [N · mm]	60.5	36.8	5.3
Force on knurled knob F_{knob} [N]	4.0	2.5	0.4

The force on the knurled knob therefore stays between 0.4 [N] and 4.0 [N] depending on the thread, which is comfortable for the operator in every case.

3.1.4 Phase II

During phase II, the force passing through the thread is applied by deforming the blade stages that separate block A from block B. The flexure guides involved in the equivalent stiffness are therefore those between the two blocks: pivot II, link II, and blade stages 1, 2 and 3, which work in parallel. Since the goal is to measure large forces while keeping a good resolution for small forces, the stiffness k_{eq} is not linear and takes different values depending on the measured force F and the stroke x:

F [N]	**x [mm]**	k_{eq} **[N/mm]**
0 – 20	0 – 5	4
20 – 50	5 – 6.5	20
50 – 100	6.5 – 7.5	50

The contribution of link II to the equivalent stiffness was calculated in the previous section and is 0.014 [N/mm]. Since this is several orders of magnitude below the first target stiffness of 4 [N/mm], link II is neglected in the following calculations. Pivot II and blade stages 1, 2 and 3 therefore provide the three different stiffnesses. They are arranged so that at the start of the stroke only stage 1 works, while stages 2 and 3 only start working once the stroke exceeds given thresholds.

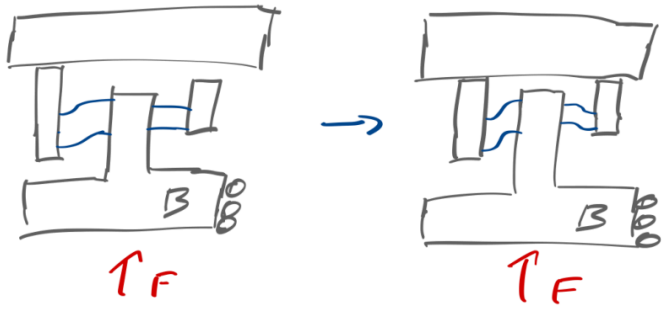


Figure 7: Blade-stage arrangement giving an equivalent stiffness that varies with the stroke

The stiffnesses of the three blade stages must therefore satisfy:

$$k_{eq,1} = k_{\text{pivot}} + k_1$$
$$k_{eq,2} = k_{\text{pivot}} + k_1 + k_2$$
$$k_{eq,3} = k_{\text{pivot}} + k_1 + k_2 + k_3$$

Since the equivalent stiffness of pivot II was calculated in the previous section, this system is easily solved, and a numerical application gives:

Stage no.	**1**	**2**	**3**
k [N/mm]	3.94	16	30

The blade stages are sized with the same formulas as the blade stage of phase I (see previous section):

Stage no.	**E [MPa]**	**l [mm]**	**b [mm]**	**h [mm]**	**k [N/mm]**
1	193000	56	14.41	0.5	3.94
2	193000	35	14.2	0.5	16
3	193000	25	9.71	0.5	30

3.1.5 Other guides

Link III and blade stage II are not part of the measuring chain. Their stiffness only affects the force needed to actuate the mechanism, so their dimensions can be chosen freely. Their stiffnesses are calculated in the same way as for phase I:

	E [MPa]	**l [mm]**	**b [mm]**	**h [mm]**	**k [N/mm]**
Blade stage II	193000	50	14.6	0.2	0.36

	E [MPa]	**l [mm]**	**b [mm]**	**h [mm]**	**r [mm]**	**k [N/mm]**
Link III	193000	45	10	0.01	15	0.06

3.2 Force-displacement characteristics and amplification

Throughout the calculations, the motion amplification is assumed to be constant.

3.2.1 Phase I

During phase I, the motion is amplified by 8. The maximum displacement of 2 [mm] at the wound therefore produces a displacement of 16 [mm] at the tip of the needle. The needle thus sweeps 20 graduations of 0.8 [mm], which gives a resolution of 0.05 [N], 0.2 [N] and 0.4 [N] for the EP3, EP6 and EP9 threads respectively. For all three threads, the mechanism can measure forces up to $\frac{1}{10}\boldsymbol{F}_{r,\mathbf{nominal}}$ with a resolution of $\frac{1}{200}\boldsymbol{F}_{r,\mathbf{nominal}}$, for a maximum wound closing of 2 [mm].

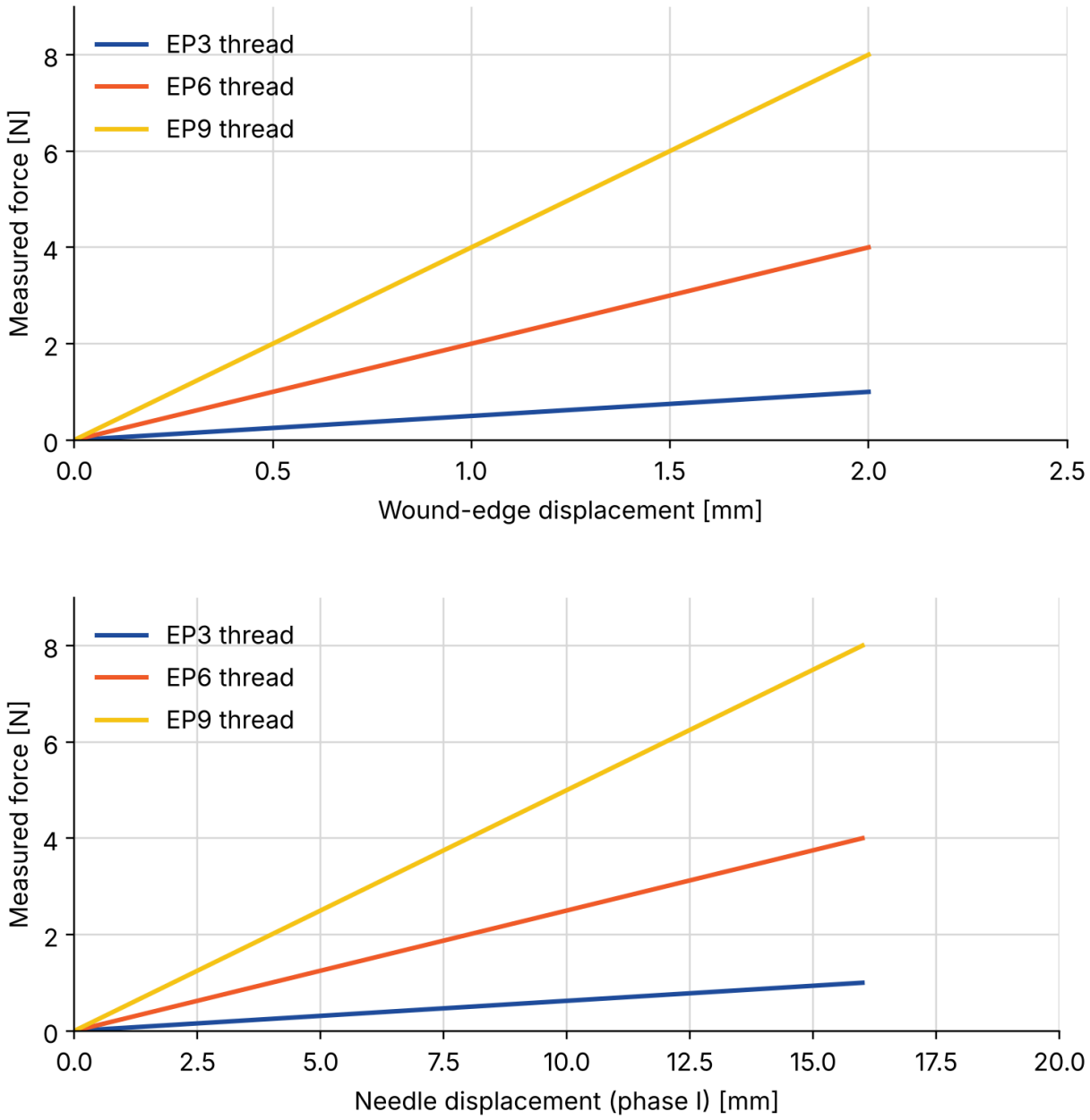


Figure 8: Measured force as a function of the wound-edge displacement (top) and of the needle displacement (bottom) during phase I

The parasitic motion λ mentioned in section 2.1 can also be calculated from the maximum stroke x_{max} and the blade length l of blade stage I, using the relation:

$$\lambda \simeq \frac{3x_{max}^2}{5l}$$

A numerical application gives:

x_{max} [mm]	l [mm]	λ [mm]
4	50	0.48

3.2.2 Phase II

During phase II, the equivalent stiffness of the system is not linear and changes with the displacement. The maximum stroke of 7.5 [mm] is amplified by 6, which gives a total displacement of 45 [mm] at the tip of the needle. The mechanism can measure a maximum force of 100 [N]. The needle sweeps 75 graduations spaced 0.6 [mm] apart over the full stroke, and the resolution remains at or below one tenth of the measured force over the whole stroke.

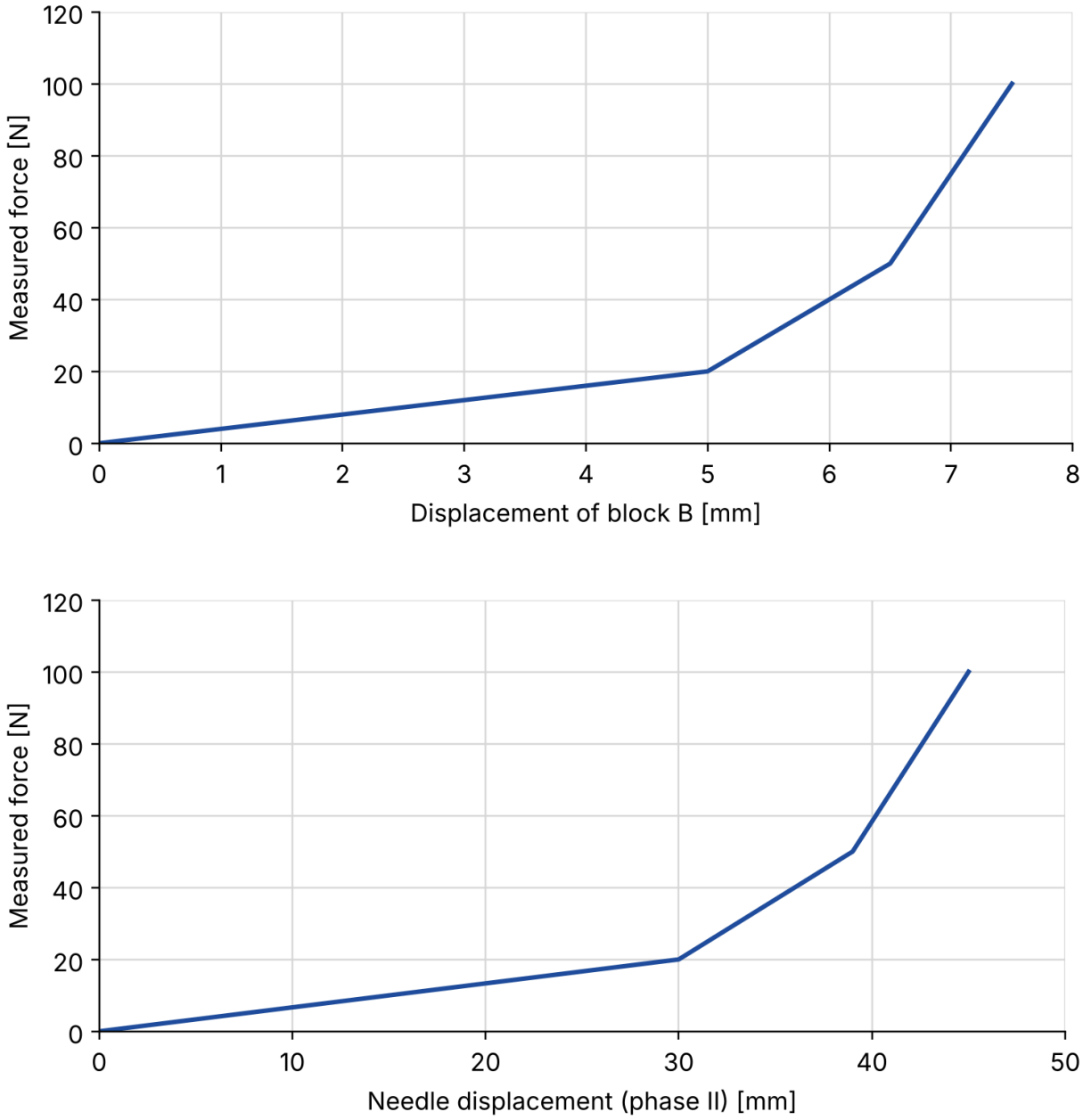


Figure 9: Measured force as a function of the displacement of block B (top) and of the needle (bottom) during phase II

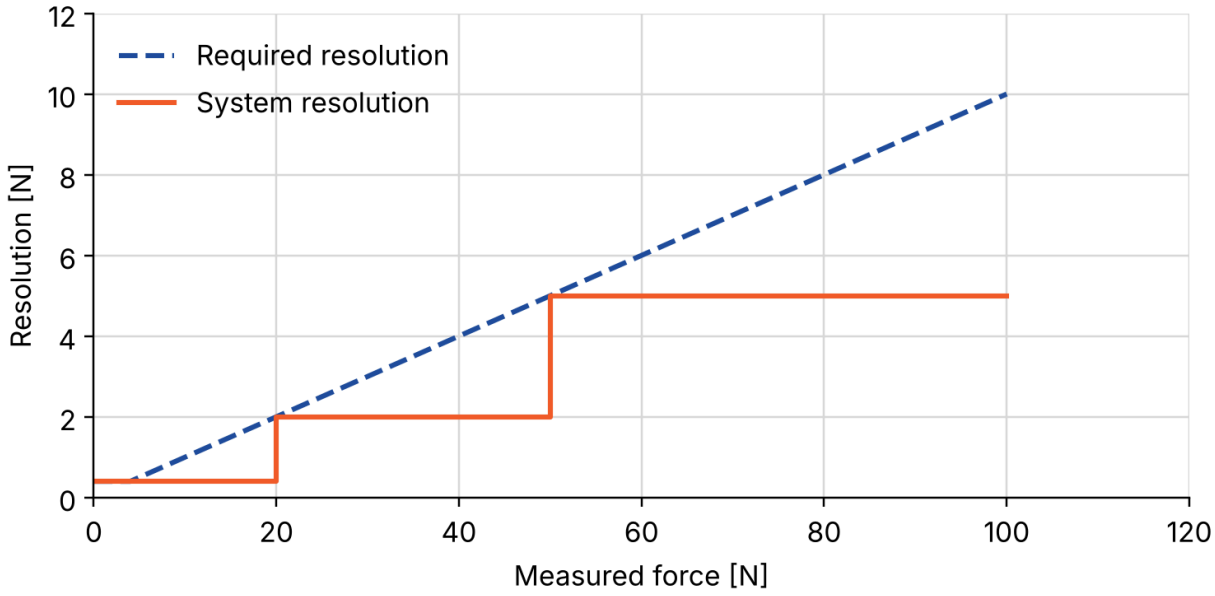


Figure 10: Resolution as a function of the measured force during phase II

3.3 Maximum stresses

3.3.1 Method

To find the maximum stresses in all the flexure elements, the maximum stroke of each element must be known. In the mechanism, the maximum strokes are set by the displacement of block A relative to the frame while the knot is tightened in phase I, and by the displacement of block B during phase II. Using the small-angle approximation $\sin(\alpha) \simeq \alpha$, the maximum angular stroke of the pivots and links is:

$$\theta_{max} \simeq \sin(\theta_{max}) = \frac{x_{max}}{r}$$

Once the maximum strokes are known, the maximum stresses can be found with the following relations[4]:

$$\sigma_{\text{pivot, link, max}} = \frac{(2hl + 3hp)\, E\, \theta_{max}}{l^2} \qquad \sigma_{\text{stage, max}} = \frac{3Ehx_{max}}{l^2}$$

Since stage I is preloaded, its maximum stress is higher than in the other blade stages and is calculated with the following formula:

$$\sigma_{\text{stage I, max}} = \frac{1}{bh}\left(\frac{x_{max}\left(3bEh^3\pi^2 + 3l^2N(\pi^2 - 12)\right)}{hl^2\pi^2} + N\right)$$

[4] Formulas from the book "Conception des guidages flexibles" [1].

3.3.2 Calculations

Blade stage I is deformed the most when block A moves by $x_{max} = 2$ [mm]. This displacement rotates link I and pivot I by 0.2 [rad]. Pivot II and link II reach their maximum angular stroke when blocks A and B have moved 9.5 [mm] closer (2 [mm] during phase I + 7.5 [mm] during phase II), which gives 0.38 [rad]. Blade stages 1, 2 and 3 have maximum displacements of 7.5 [mm], 1.5 [mm] and 1 [mm] respectively. Finally, blade stage II has a maximum displacement of 13.5 [mm] (9.5 [mm] from phases I and II + 4 [mm] of possible relaxation), which corresponds to a maximum angular stroke of 0.9 [rad] for link III. The maximum stresses are then:

	E [MPa]	**h [mm]**	**l [mm]**	x_{max} **[mm]**	σ_{max} **[MPa]**
Stage II	193000	0.2	50	13.5	625.3
Stage 1	193000	0.5	56	7.5	693.4
Stage 2	193000	0.5	35	1.5	354.9
Stage 3	193000	0.5	25	1	463.2

	E [MPa]	**h [mm]**	**l [mm]**	**p [mm]**	θ_{max} **[rad]**	σ_{max} **[MPa]**
Pivot I	193000	0.1	40	10	0.2	441.5
Link I	72000	0.1	40	0	0.2	74.7
Pivot II	193000	0.1	40	10	0.38	504.2
Link II	193000	0.1	40	0	0.38	366.7
Link III	193000	0.1	45	0	0.9	694.8

	E [MPa]	**h [mm]**	**l [mm]**	**b [mm]**	**N [N]**	x_{max} **[mm]**	σ_{max} **[MPa]**
Stage I	193000	0.5	50	10	157.8	4	331.26

The allowable stress is 735 [MPa] for all parts (except link I, which is made of aluminium and has an allowable stress of 110 [MPa]), so the reliability of the mechanism is guaranteed for 10 million cycles.

3.4 Sizing of the lever

The lever is a bar of circular cross-section whose pivot splits it into an input arm $l_1 = 180$ [mm] and an output arm $l_2 = 15$ [mm]. The force applied by the operator is $F_{in} = 10$ [N].

The lever ratio gives the force amplification:

$$i = \frac{l_1}{l_2} = \frac{180}{15} = 12$$

The force transmitted to the mechanism at the output is therefore:

$$F_{out} = i \cdot F_{in} = 12 \times 10 = 120 \text{ [N]}$$

3.4.1 Reaction at the pivot

The plain bearing at the lever pivot carries both the input force and the reaction of the mechanism. Since these two forces act in the same direction on the pivot, the resulting reaction is:

$$R = F_{in} + F_{out} = 10 + 120 = 130 \text{ [N]}$$

This value $R = 130$ [N] is used to size the bearing, as it is the worst load case.

3.4.2 Contact pressure check

The plain bearing is cylindrical, with a shaft diameter $d = 13$ [mm] and a length $L = 20$ [mm]. The mean contact pressure is:

$$p = \frac{R}{d \cdot L} = \frac{130}{13 \times 20} = 0.5 \text{ [MPa]}$$

The bearing material is a tin bronze, with an allowable pressure $p_{adm} = 7$ [MPa]. The condition is satisfied:

$$p = 0.5 \text{ [MPa]} \ll p_{adm} = 7 \text{ [MPa]}$$

The contact pressure requirement is therefore met.

3.5 Sources of measurement error

The value shown by the phase II needle is not the force passing through the thread. The force measured by the mechanism during phase II is the force added once phase I is over. To know the total force passing through the thread, the values displayed at the end of phases I and II must be added together. It is important to read the phase I value before starting phase II, to avoid any error caused by relaxation of the wound.

In addition, if the wound closes by less than 2 [mm] during phase I, this causes a small error in phase II. Ideally, the first of the three phase II blade stages starts to deform as soon as phase II begins. Depending on the knot force in phase I, however, blocks A and B may have to be brought closer before phase II starts, which deforms link II and pivot II. The deformation of these two flexure elements introduces a small error, in the form of an additional force passing through the thread without being measured.

The motion amplification is also assumed to be constant, which is not true: it varies with the stroke, and these variations are not taken into account in the display. One solution would be to calculate the variation of the amplification and include it in the graduation of the scale.

Finally, some angular strokes in the mechanism (for example for link III) are large and may fall outside the range where the small-angle approximation applies. The corresponding calculations may therefore contain errors. One solution would be to use a real link instead of a bending blade.

4 Discussion

The dimensions of our mechanism exceed those set by the specifications (26.15 × 23 × 6.49 cm). Since the blades in our mechanism are several centimetres long, it is difficult to arrange the elements compactly. This also increases the weight of the mechanism. It slightly reduces portability compared with a mechanism that fully meets the maximum envelope (25 × 25 × 5 cm), which would make transport and storage harder if the device had to be moved often (in a crisis situation, for example). Other blade arrangements for the flexure elements (cross-blade pivots instead of RCC pivots) could reach similar stiffnesses with smaller dimensions. The maximum blade thickness used in the mechanism is limited to 0.1 [mm], but even thinner blades could be used. This would of course reduce the size of the blades, and also reduce the preload needed on stage I. The preload spring would then not need to be as wide and as long, which would make the mechanism even smaller.

5 Construction

All flexure elements are made of steel blades, except link I, which is made of aluminium to lower its stiffness. The blades are attached to the frame and to the blocks with screws and with dedicated parts that position and align them. The other parts are made of aluminium, of steel where stresses are high (all parts in direct contact with the blades, to keep a constant effective length by preventing indentation of the contact surfaces), and of plastic otherwise. There is also a Mädler 62353222 bronze plain bearing (10 mm bore). The plastic parts can be 3D printed. The elements clamping the blades to the frame are made of steel to prevent indentation and keep a constant, effective clamping force. We used ISO 4762 hex socket screws, reusing the same screws (diameter and length) as much as possible. The two circlips were modelled as washers with a 9.5 mm inner diameter and 1 mm thickness, to fit the 10 mm shaft carrying the lever. The key is a standard 3 × 10 mm key.

The top plate is made of acrylic so that the surgeon can admire our mechanism and read the tightening force (Figure 14). Our scale is made of stickers (Figures 15 and 16), which is why it does not appear in the assembly drawings. For phase I, it is a 16 mm arc (80 mm radius) with 20 graduations of 0.05 N. For phase II, it is a 45 mm arc (150 mm radius) with 75 graduations of progressively 0.4 N (0–20 N), 2 N (20–50 N) and 5 N (50–100 N).

The stiffness is adjusted for the thread type with an Allen key, inserted through one of the vertical walls. Notches on the rails show how far the preload screw must travel, so the surgeon does not have to count turns (Figure 18). The thread is tested with the black lever (Figure 19). The added grip is crucial, because it makes the lever easy to hold: in a conflict zone, sweat, blood or dust could otherwise make the test impossible. We had designed a compartment under the lever to store the material needed for the procedure as well as the Allen key (Figure 20), closed by a magnetic lid. This idea does not appear in the final design, since the specifications require no removable parts. With more time, a non-removable drawer could have been designed.

The small tab sticking out of the top plate is used to reset the drag needle. We built the StitchMeter trying to make the parts as machinable as possible and reusing parts as often as possible. This was a trade-off with the size of the housing. Unlike the rest of the mechanism, we did not have time to split the housing (backplate) into several parts, so it is a single piece. It could be made by bending, but the best solution would be to screw the vertical walls onto support pillars. The top plate is not fastened either; ideally it would be screwed onto these pillars.

Assembly sequence:

1. Assemble all the blade stages (Figure 21).
2. Assemble the two pivots (Figure 22).
3. Assemble the lever sub-assembly (Figure 23).
4. First step of the final assembly: attach the mobile wound-edge holder to its support (Skinbati-right), then attach its blade stage, as well as the spring carriage (with its long screw and guide shafts), and fix everything to the bottom plate (in the images, this plate is a version created only for rendering, hence without holes). The spring can be placed between the carriage and its support on the bottom plate. The fixed wound-edge holder can also be attached to the plate (Figure 24).
5. Second step of the final assembly: attach the Skinbati-right extension, which will later hold the pivot of the phase II needle, and place the pre-assembled phase I needle pivot (Figure 25).
6. Third step of the final assembly: assemble the three blade stages that pull on the thread in phase II with their pushers and their support plate, as well as the blade stage linking this plate to the frame. Everything can then be attached to the frame (Figure 26).
7. Fourth step of the final assembly: place the phase II pivot and the lever sub-assembly. The side plates can then be attached (had we modelled them that way) (Figure 27).
8. Press the two halves of the drag needle together, with the top plate between them (the needle is made of plastic so it can be assembled with a simple press fit) (Figure 28). Finally, add the top plate and apply the scale stickers.

6 Conclusion

The mechanism gives an accurate measurement of the thread tension, with a very good resolution during phase I. During phase II, the mechanism measures the forces with the required resolution and good readability at the needle, thanks to the evenly spaced graduations. However, the device remains bulky, with large strokes that may introduce approximation errors. To address these issues, some elements could be resized to reduce the strokes and make the mechanism more compact. Small monolithic flexure elements could also be considered if cost allows. This version of the mechanism therefore leaves many possible improvements, both in size and in performance.

7 Appendices

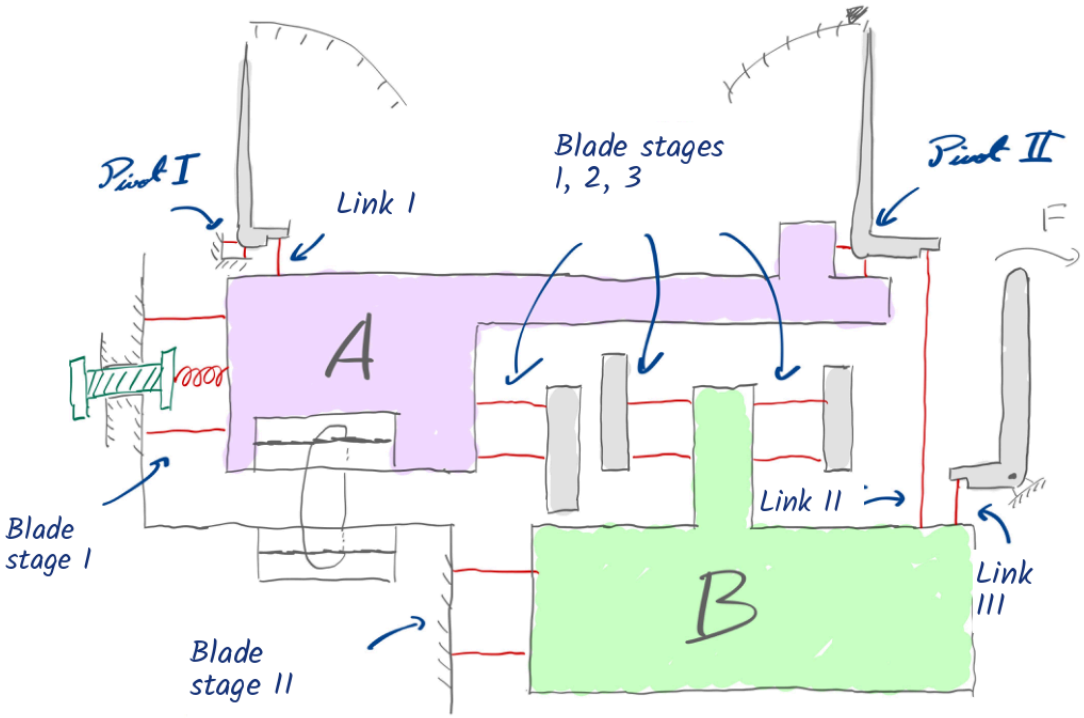


Figure 11: Flexure element references

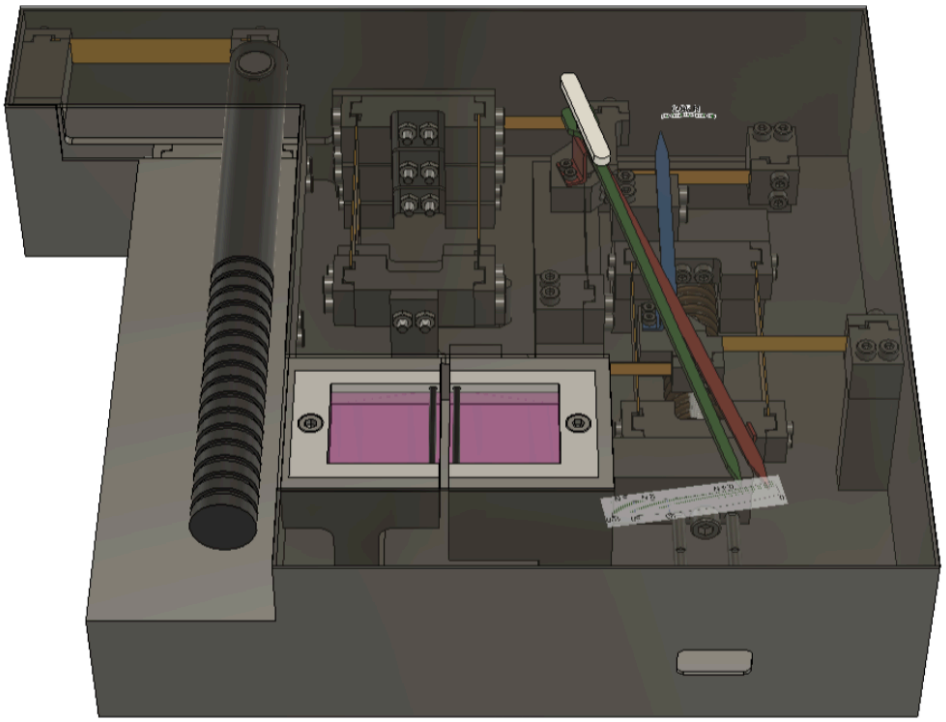


Figure 12: View of the complete housing

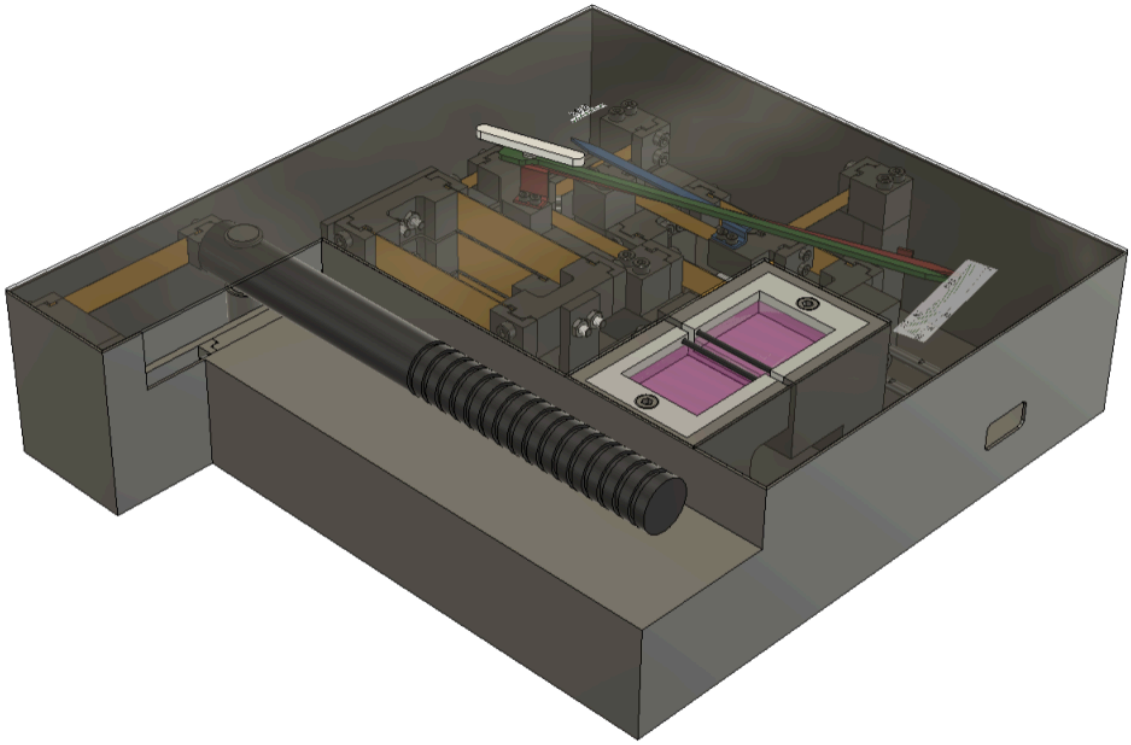


Figure 13: View of the complete housing

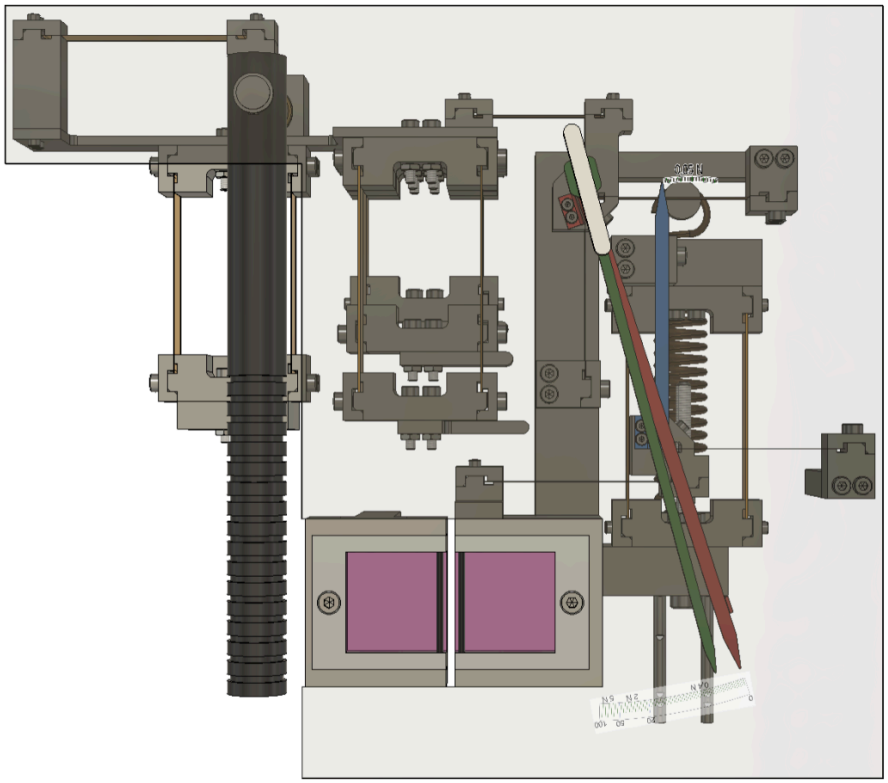


Figure 14: Top view of the complete housing

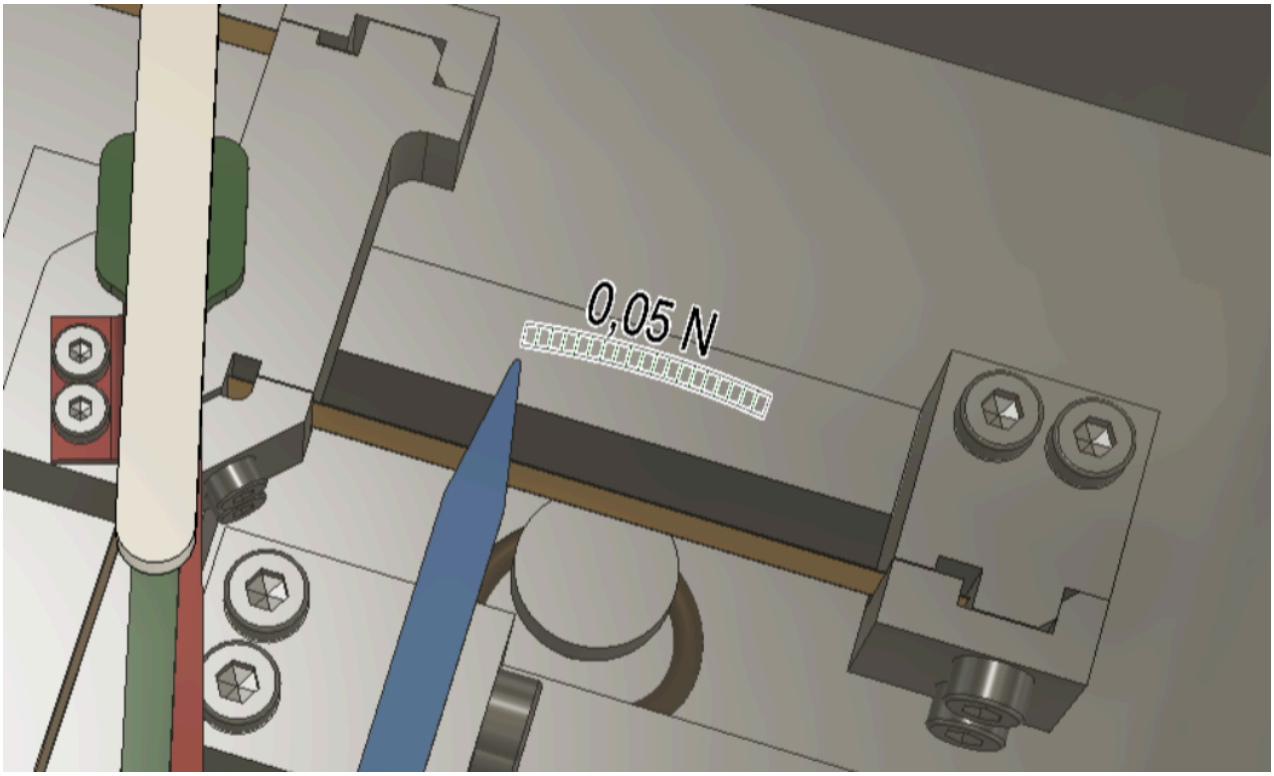


Figure 15: Scale sticker for phase I

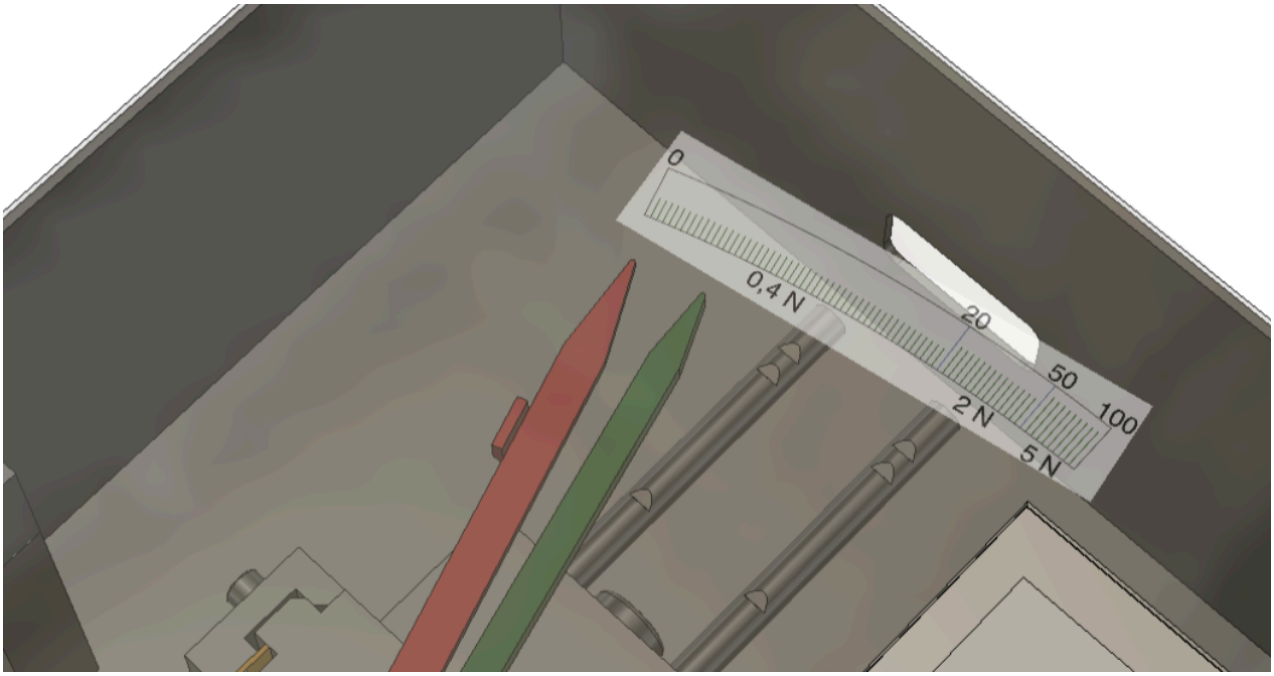


Figure 16: Scale sticker for phase II, with the drag needle

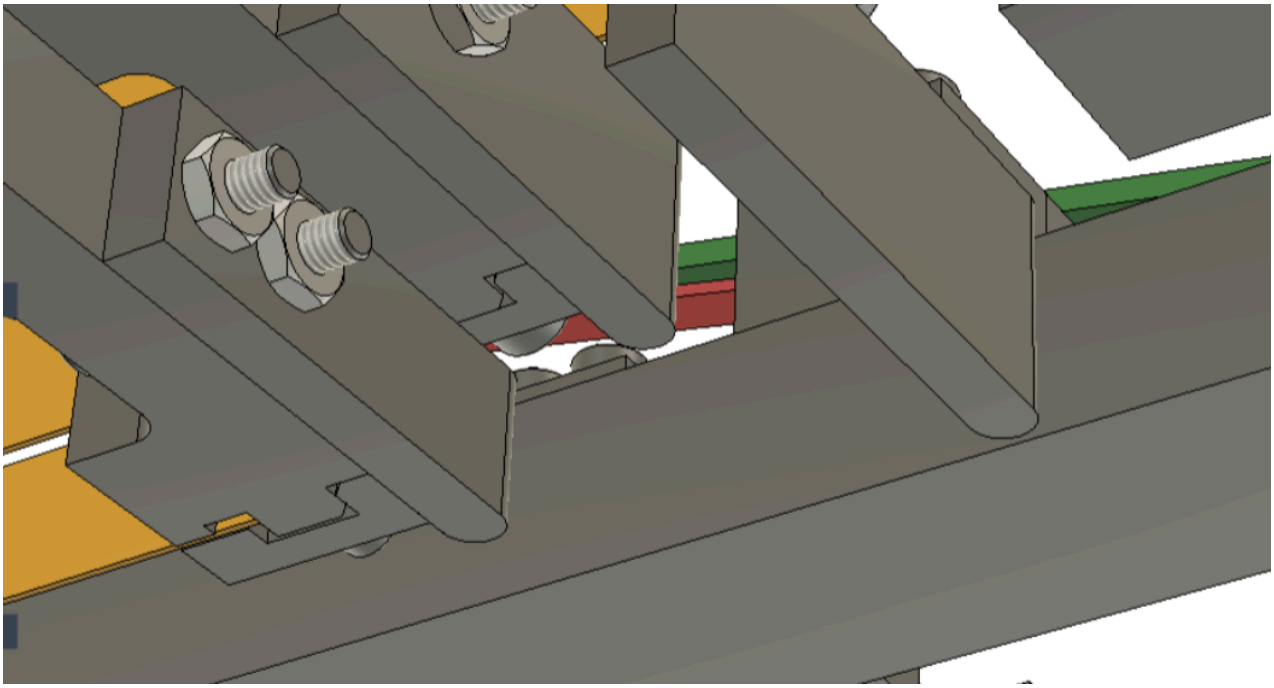


Figure 17: The non-linear spring made of the 3 blade stages

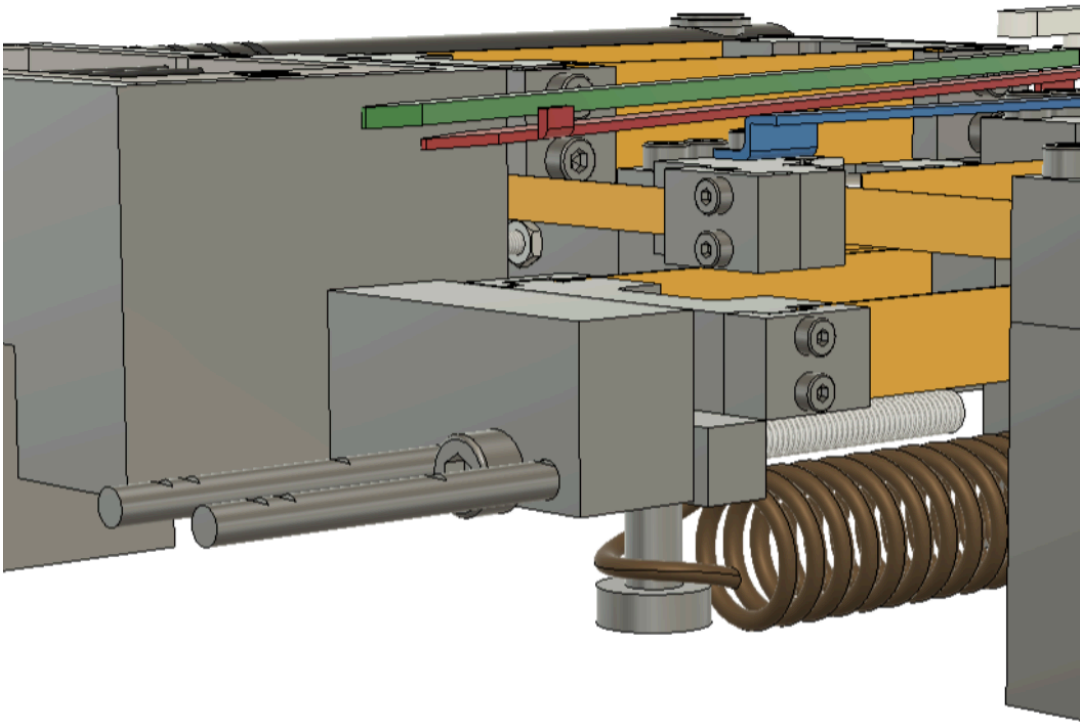


Figure 18: Preload screw, rails with notches

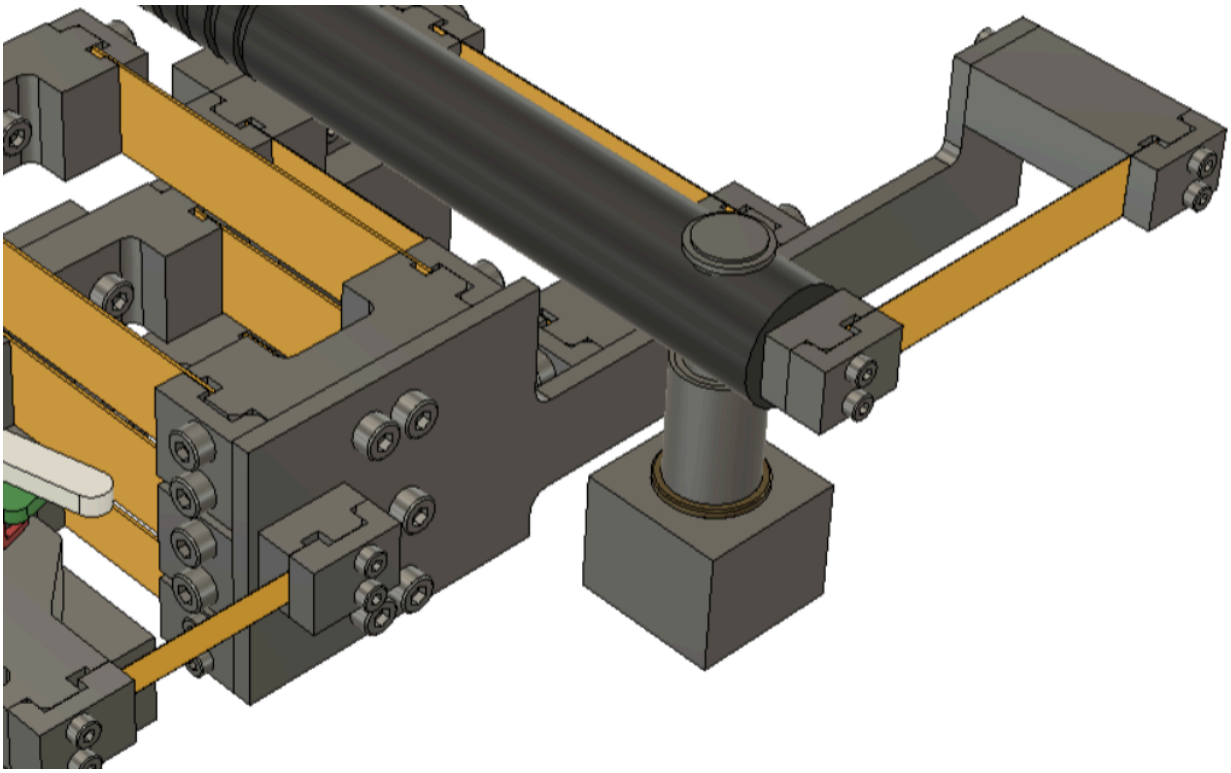


Figure 19: Lever and plain bearing

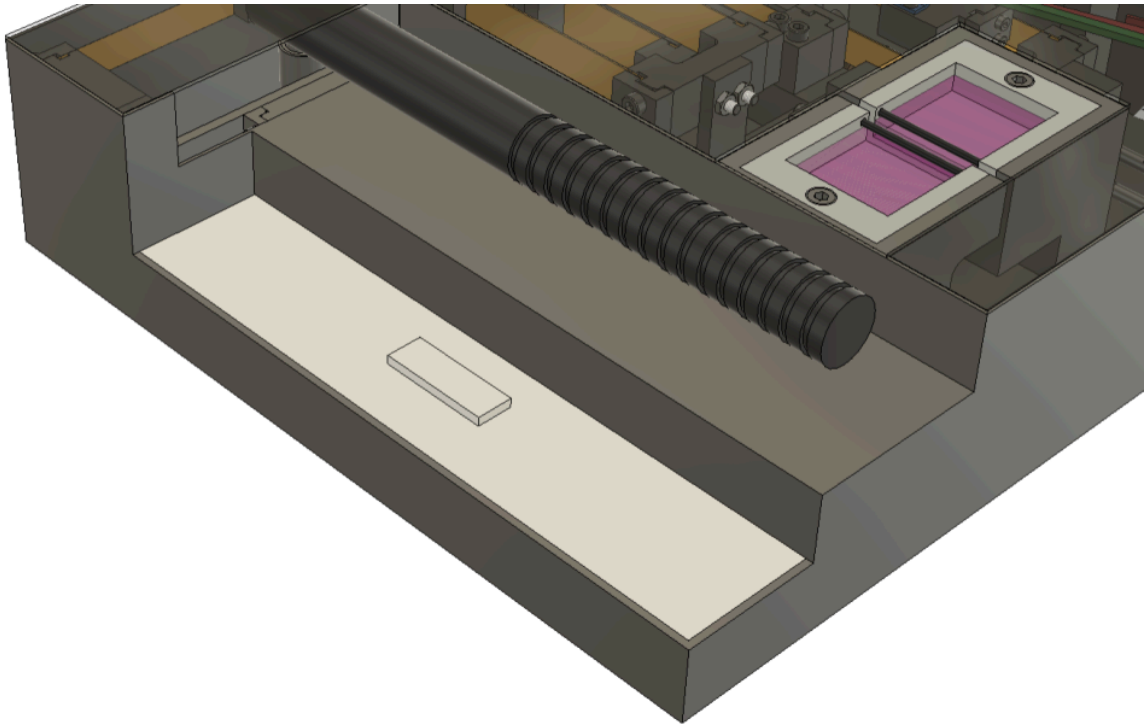


Figure 20: Storage compartment (not included in the final design)

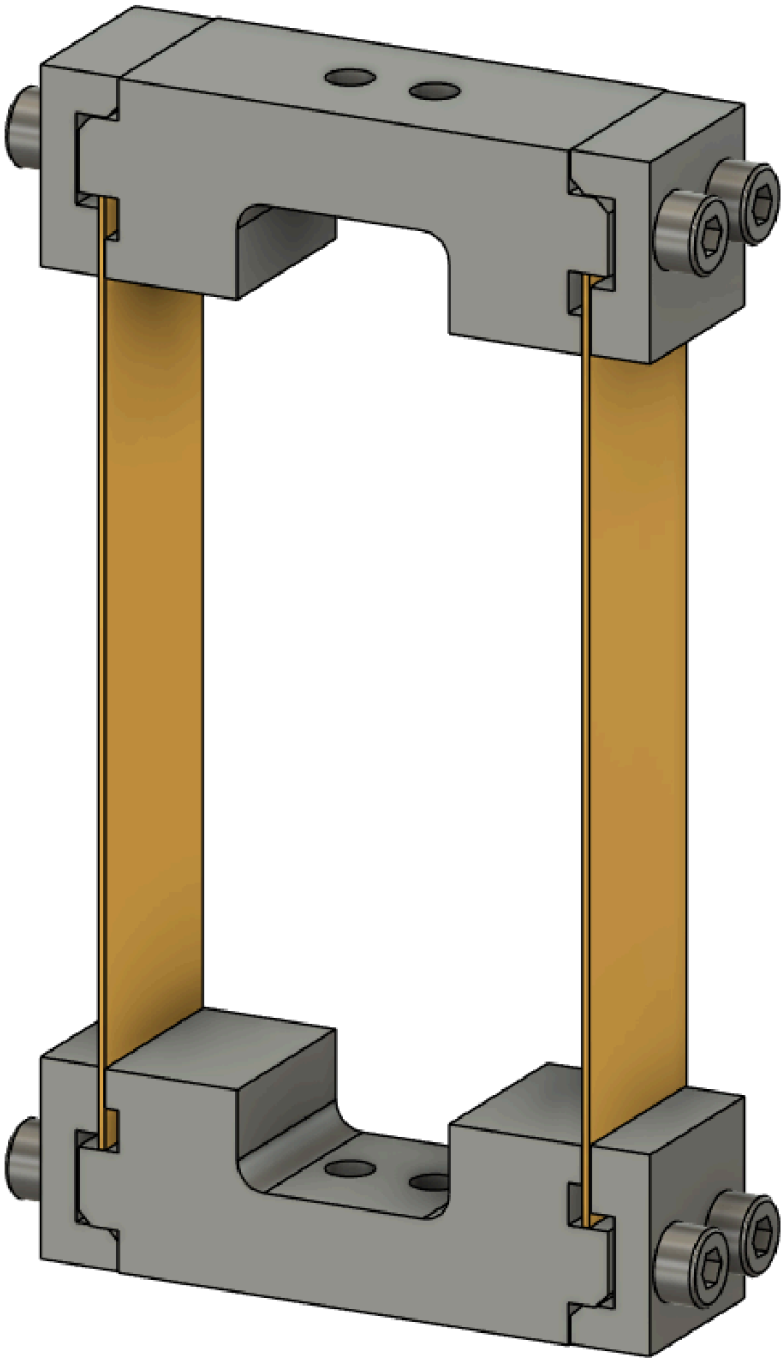


Figure 21: Blade stage assembly

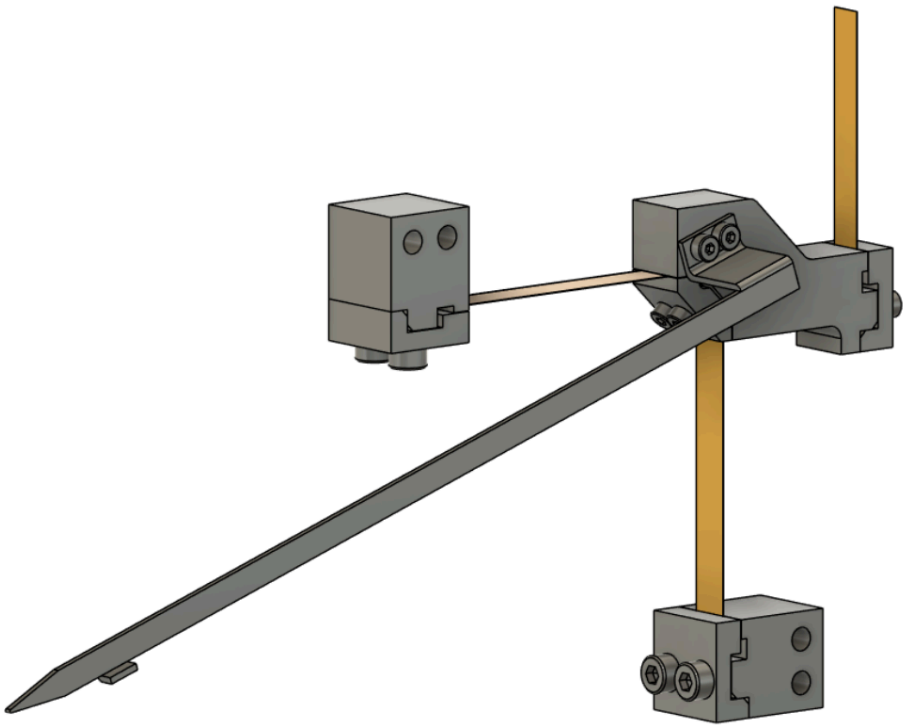


Figure 22: Pivot assembly

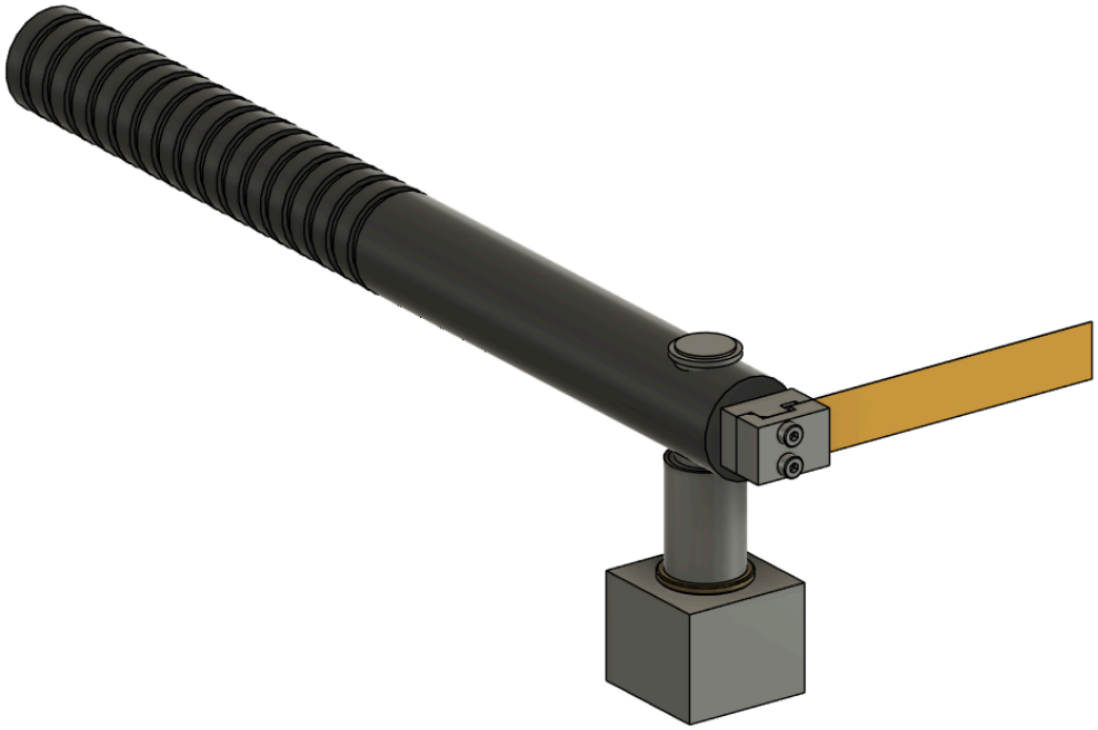


Figure 23: Lever sub-assembly

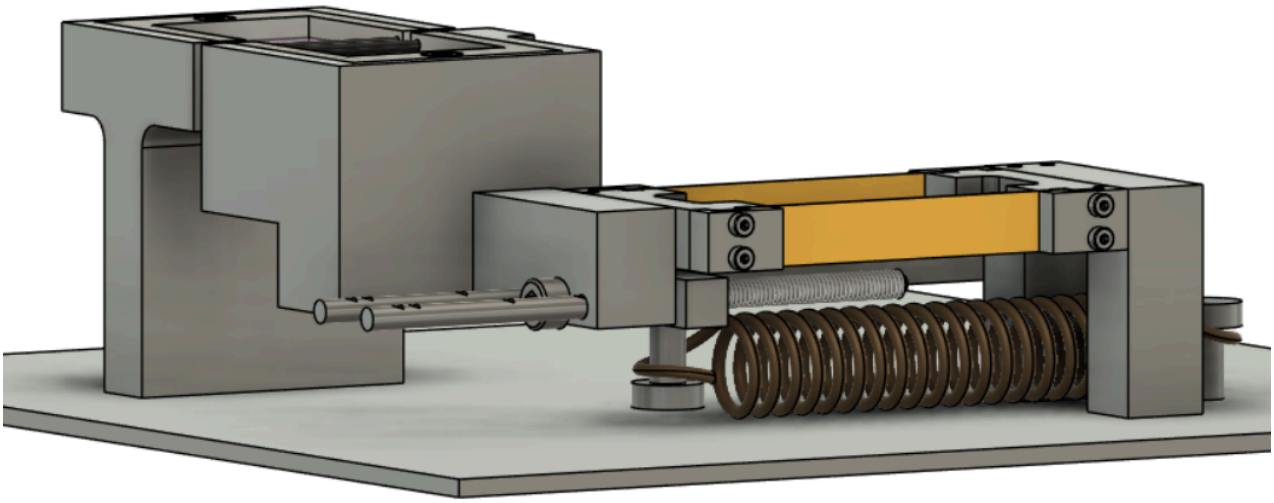


Figure 24: Final assembly, step 1

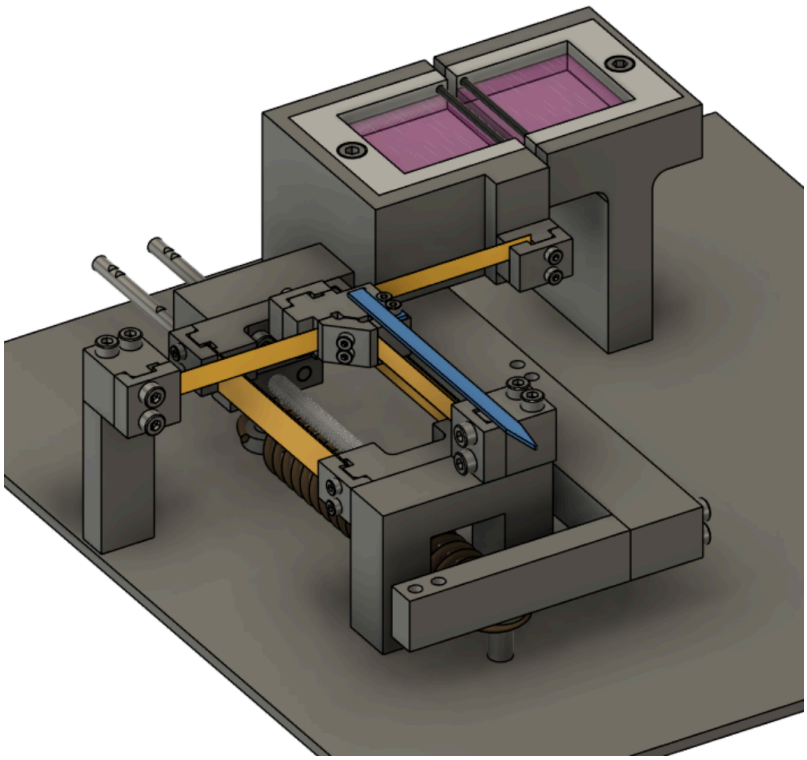


Figure 25: Final assembly, step 2

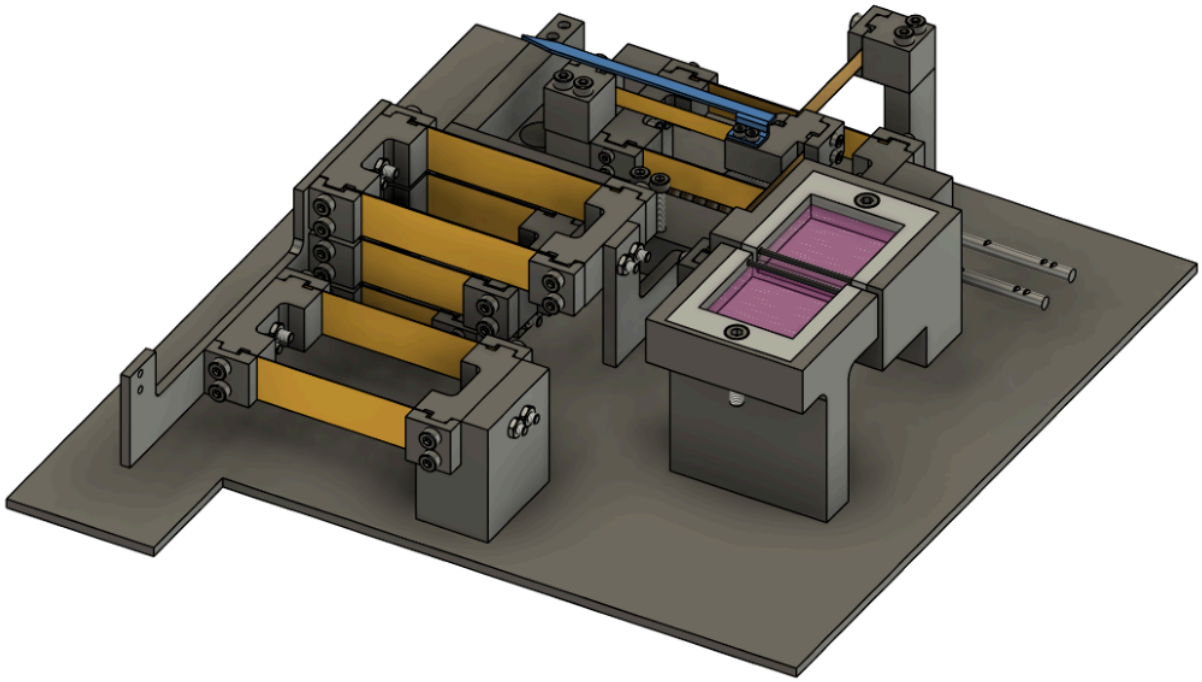


Figure 26: Final assembly, step 3

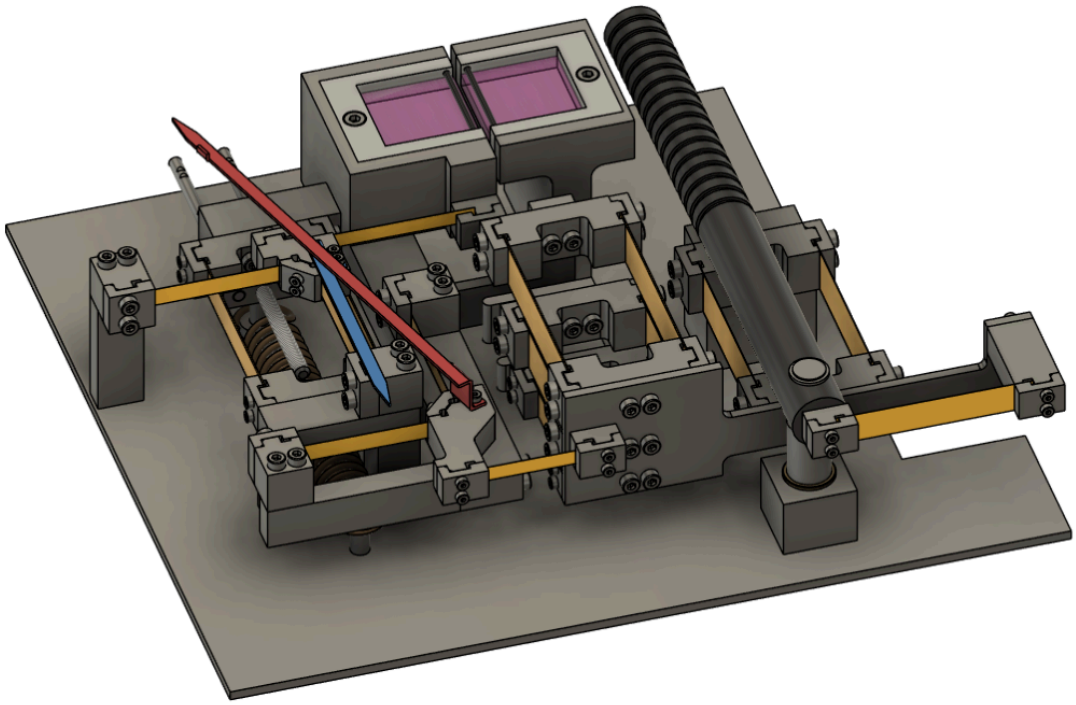


Figure 27: Final assembly, step 4

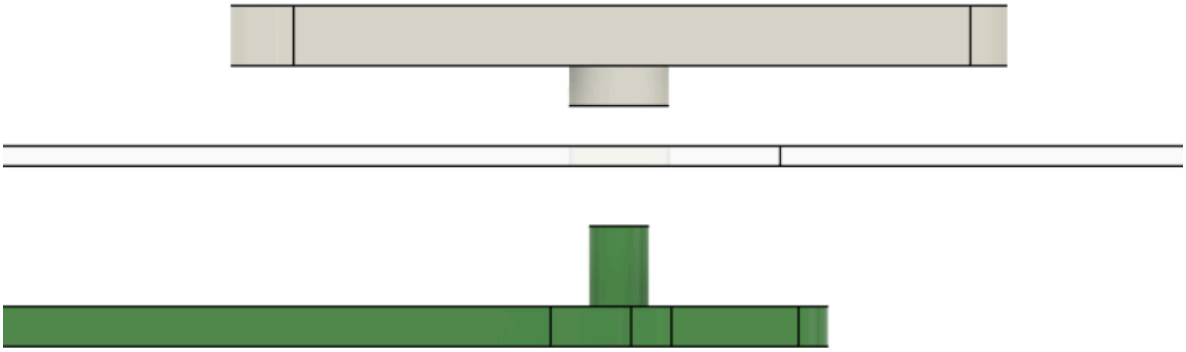
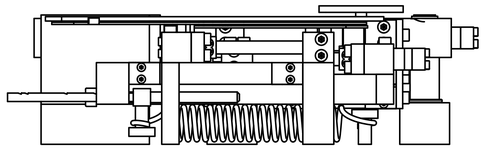
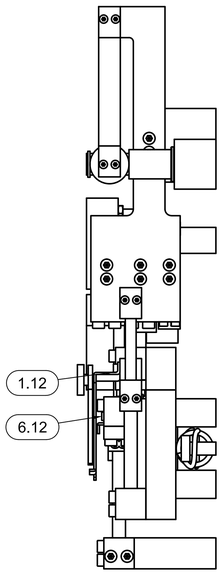


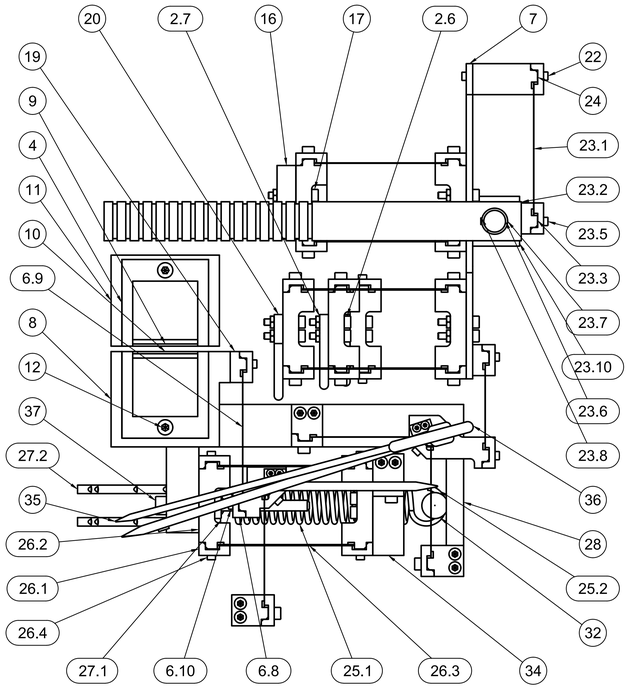
Figure 28: Fitting the drag needle to the top plate

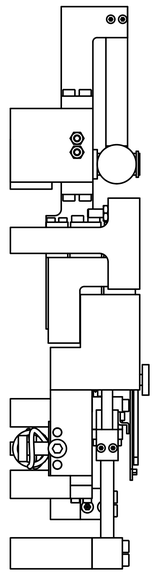
References

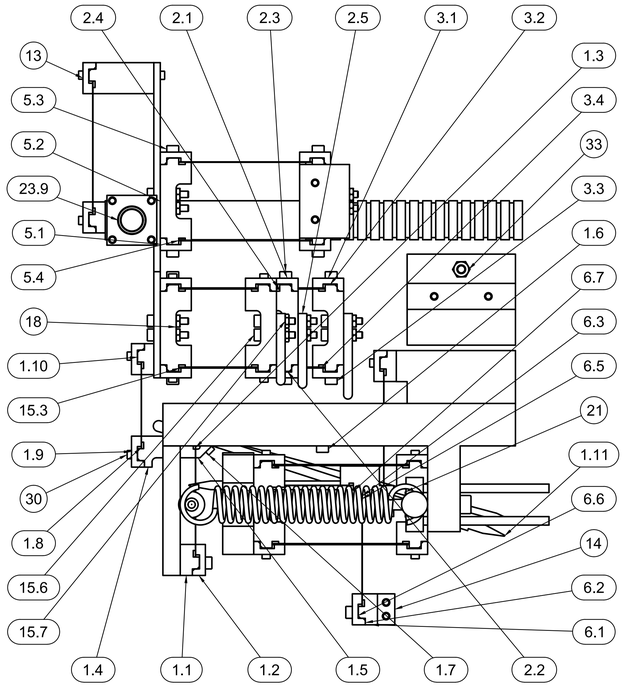
[1] Simon Henein. *Conception des guidages flexibles*. Presses Polytechniques et Universitaires Romandes.

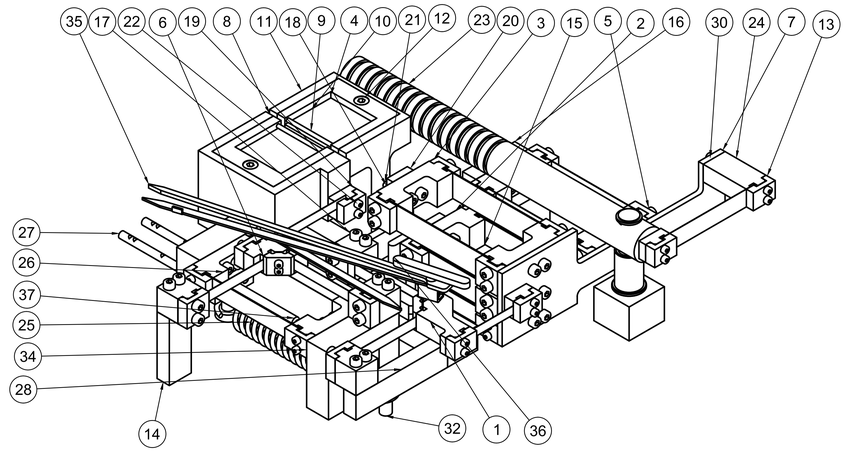












Parts List			
Item	Qty	Part Number	Material
1	1	Hand_asb_P2	Steel
1.1	2	BH_pivot_14.6	Steel
1.2	2	BH_small_14.6_0.1	Steel
1.3	2	Blade_P	Steel
1.4	1	Pivot_P2	Steel
1.5	1	BH_angled_14.6	Steel
1.6	4	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
1.7	2	DIN EN ISO 4762 - M2 x 8 Steel 4.6 Plain	Steel 4.6, Plain
1.8	1	Blade_B2	Steel
1.9	2	BH_small_14.6_0.1_pivot	Steel
1.10	1	BH_Bielle	Steel
1.11	1	Aiguille_P2	Steel
1.12	2	DIN EN ISO 4762 - M2 x 6 Steel 4.6 Plain	Steel 4.6, Plain
2	1	BH_asb_T2	----
2.1	4	BH_small_14.6_0.1	Steel
2.2	2	BH_14.6	Steel
2.3	8	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
2.4	2	Blade_13.72	Steel
2.5	1	Pusher_T2	Steel
2.6	2	DIN EN ISO 4762 - M3 x 16 Steel 4.6 Plain	Steel 4.6, Plain

Parts List			
Item	Qty	Part Number	Material
2.7	2	DIN 439-2 - M3 x 0.5 Steel 6 Plain	Steel 6, Plain
3	1	BH_asb_T1	Steel
3.1	4	BH_small_14.6_0.1	Steel
3.2	2	BH_14.6	Steel
3.3	8	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
3.4	2	Blade_T1	Steel
4	2	Skinholder	Steel
5	1	BH_asb_TII	Steel
5.1	4	BH_small_14.6_0.2	Steel
5.2	2	BH_14.6	Steel
5.3	8	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
5.4	2	Blade_TII	Steel
6	1	Hand_asb_P1	----
6.1	2	BH_pivot_14.6	Steel
6.2	2	BH_small_14.6_0.1	Steel
6.3	2	Blade_P	Steel
6.4	1	Hand_80	Steel
6.5	1	BH_angled_14.6	Steel
6.6	4	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
6.7	2	DIN EN ISO 4762 - M2 x 8 Steel 4.6 Plain	Steel 4.6, Plain

Parts List			
Item	Qty	Part Number	Material
6.8	1	BH_small_14.6_0.1_pivot	Steel
6.9	1	Blade_B1	Aluminum
6.10	2	DIN EN ISO 4762 - M2 x 10 Steel 4.6 Plain	Steel 4.6, Plain
6.11	1	Aiguille_P1	Steel
6.12	2	DIN EN ISO 4762 - M2 x 6 Steel 4.6 Plain	Steel 4.6, Plain
7	1	Plaque_Tables	Aluminum
8	1	Skinbati_right	Aluminum
9	2	Stringholder	Steel
10	2	Skin	Steel
11	1	Skinbati_left	Steel
12	2	DIN EN ISO 4762 - M4 x 20 Steel 4.6 Plain	Steel 4.6, Plain
13	2	BH_small_14.6_0.1_pivot	Steel
14	1	Pivotbati_P1	Steel
15	1	BH_asb_T3	----
15.1	4	BH_small_14.6_0.1_pivot	Steel
15.2	2	BH_10	Steel
15.3	2	Blade_T3	Steel
15.4	8	DIN EN ISO 4762 - M2 x 10 Steel 4.6 Plain	Steel 4.6, Plain
15.5	1	Pusher_T3	Steel

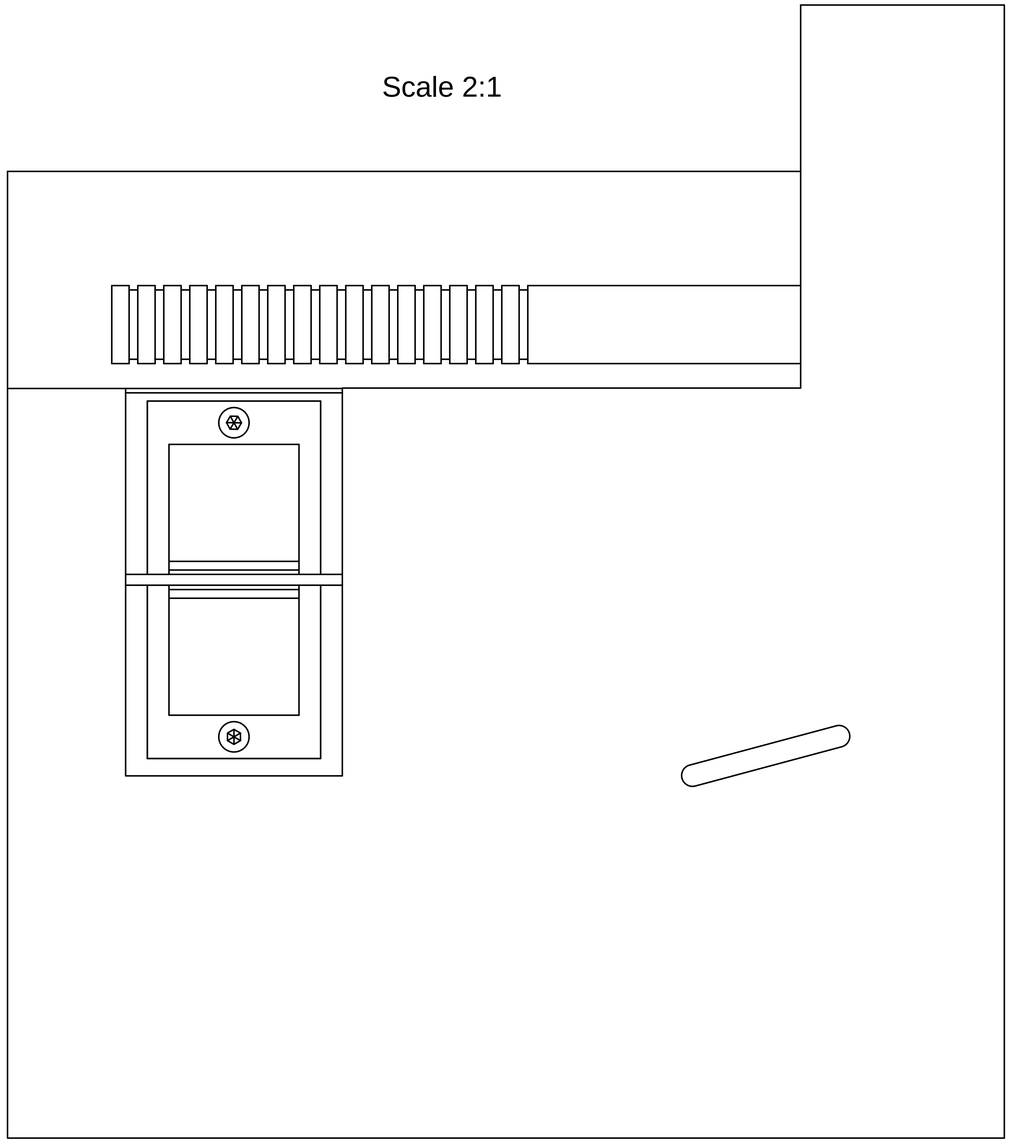
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Item	Qty	Part Number	Material
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15.7	2	DIN 439-2 - M3 x 0.5 Steel 6 Plain	Steel 6, Plain
16	1	Bati3tables	Steel
17	10	DIN EN ISO 4762 - M3 x 20 Steel 4.6 Plain	Steel 4.6, Plain
18	12	DIN 439-2 - M3 x 0.5 Steel 6 Plain	Steel 6, Plain
19	1	BH_Bielle	Steel
20	1	Pusher_T1	Steel
21	14	DIN EN ISO 4762 - M3 x 16 Steel 4.6 Plain	Steel 4.6, Plain
22	6	DIN EN ISO 4762 - M2 x 12 Steel 4.6 Plain	Steel 4.6, Plain
23	1	Levier_asb	----
23.1	1	Blade_B3	Steel
23.2	1	Lever	Aluminum
23.3	1	BH_Bielle	Steel
23.4	1	BH_small_14.6_0.1_pivot	Steel
23.5	2	DIN EN ISO 4762 - M2 x 12 Steel 4.6 Plain	Steel 4.6, Plain
23.6	1	Arbre_levier	Steel
23.7	2	Circlip	Steel
23.8	1	Key	Steel
23.9	1	62351700	Bronze

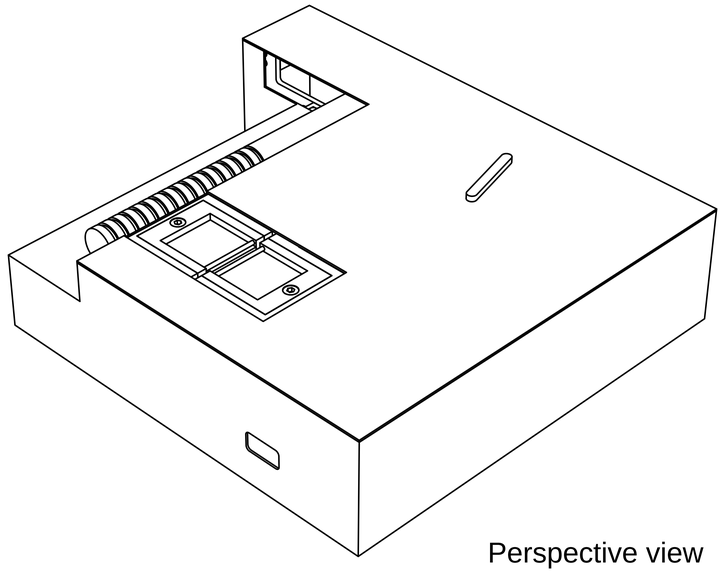
Parts List			
Item	Qty	Part Number	Material
23.10	1	Bearing support	Steel
24	1	BH_Bielle_lever	Steel
25	1	Ressort_Tension	Steel
25.1	1	Ressort_Partie_Centrale	Steel
25.2	2	Ressort_Hook	Steel
26	1	BH_asb_TI	Steel
26.1	4	BH_small_14.6_0.1_pivot	Steel
26.2	2	BH_10	Steel
26.3	2	Blade_TI	Steel
26.4	8	DIN EN ISO 4762 - M2 x 10 Steel 4.6 Plain	Steel 4.6, Plain
27	1	Chariot_Resssort_Asb	Steel
27.1	1	Chariot_Ressort	Steel
27.2	2	Spring guide	Steel
28	1	Skinbati_extension	Steel
30	4	DIN EN ISO 4762 - M2 x 8 Steel 4.6 Plain	Steel 4.6, Plain
32	1	Pin_Ressort	Steel
33	1	DIN 439-2 - M4 x 0.7 Steel 6 Plain	Steel 6, Plain
34	1	Bridge	Steel
35	1	Aiguille_Trainante	ABS Plastic
36	1	Aiguille_Trainante_Support	ABS Plastic
37	1	Vis_M5_L66	Steel

Scale 1:1

Dept.	Technical reference	Created by: Dimitri Fahmy 15/05/2026	Approved by
		Document type	Document status
EPFL		Title: Stitchmeter Internal mechanism	DWG No.
			Rev. / Date of issue / Sheet 1/2

Scale 2:1

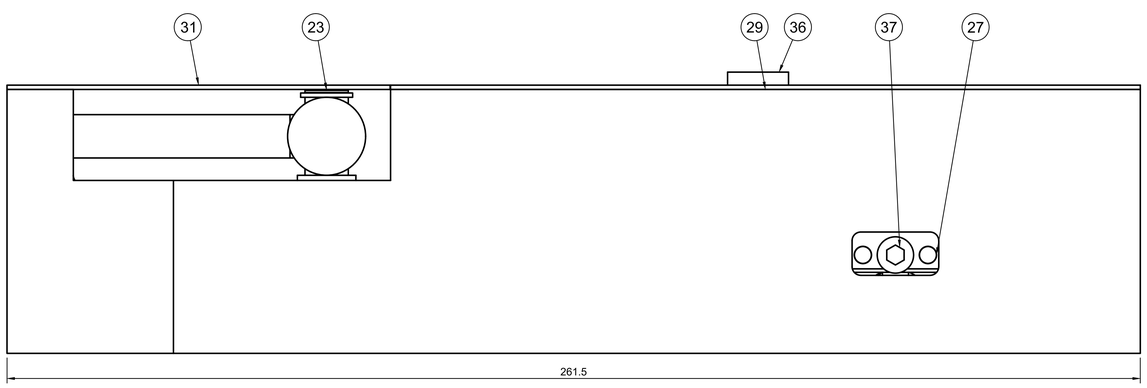




Perspective view

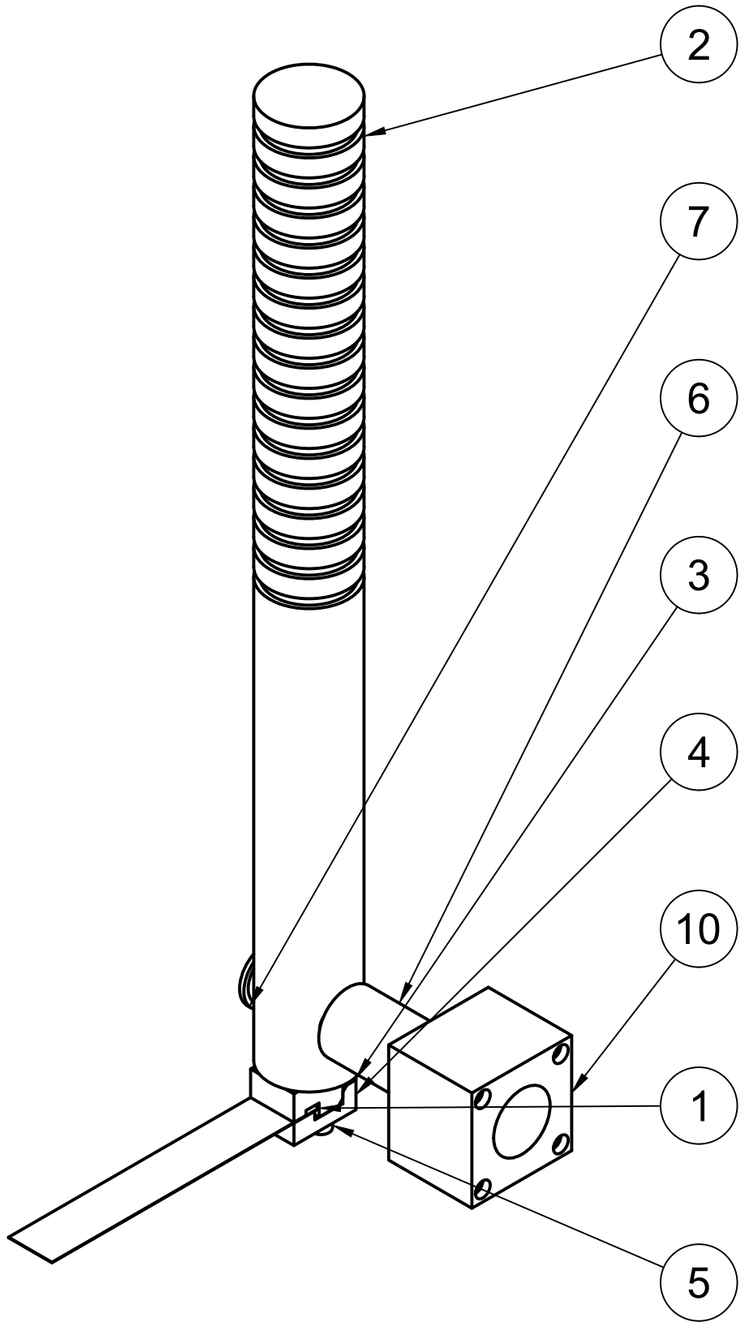
Parts List			
Item	Qty	Part Number	Material
1	1	Hand_asb_P2	Steel
2	1	BH_asb_T2	----
3	1	BH_asb_T1	Steel
4	2	Skinholder	Steel
5	1	BH_asb_TII	Steel
6	1	Hand_asb_P1	----
7	1	Plaque_Tables	Aluminum
8	1	Skinbati_right	Aluminum
9	2	Stringholder	Steel
10	2	Skin	Steel
11	1	Skinbati_left	Steel
12	2	DIN EN ISO 4762 - M4 x 20 Steel 4.6 Plain	Steel 4.6, Plain
13	2	BH_small_14.6_0.1_pivot	Steel
14	1	Pivotbati_P1	Steel
15	1	BH_asb_T3	----
16	1	Bati3tables	Steel
17	10	DIN EN ISO 4762 - M3 x 20 Steel 4.6 Plain	Steel 4.6, Plain
18	12	DIN 439-2 - M3 x 0.5 Steel 6 Plain	Steel 6, Plain
19	1	BH_Bielle	Steel
20	1	Pusher_T1	Steel
21	14	DIN EN ISO 4762 - M3 x 16 Steel 4.6 Plain	Steel 4.6, Plain
22	6	DIN EN ISO 4762 - M2 x 12 Steel 4.6 Plain	Steel 4.6, Plain
23	1	Levier_asb	----
24	1	BH_Bielle_lever	Steel
25	1	Ressort_Tension	Steel
26	1	BH_asb_TI	Steel
27	1	Chariot_Resssort_Asb	Steel
28	1	Skinbati_extension	Steel
29	1	Backplate	ABS Plastic
30	4	DIN EN ISO 4762 - M2 x 8 Steel 4.6 Plain	Steel 4.6, Plain
31	1	Topplate	Acrylic, Clear
32	1	Pin_Ressort	Steel
33	1	DIN 439-2 - M4 x 0.7 Steel 6 Plain	Steel 6, Plain
34	1	Bridge	Steel
35	1	Aiguille_Trainante	ABS Plastic
36	1	Aiguille_Trainante_Support	ABS Plastic
37	1	Vis_M5_L66	Steel

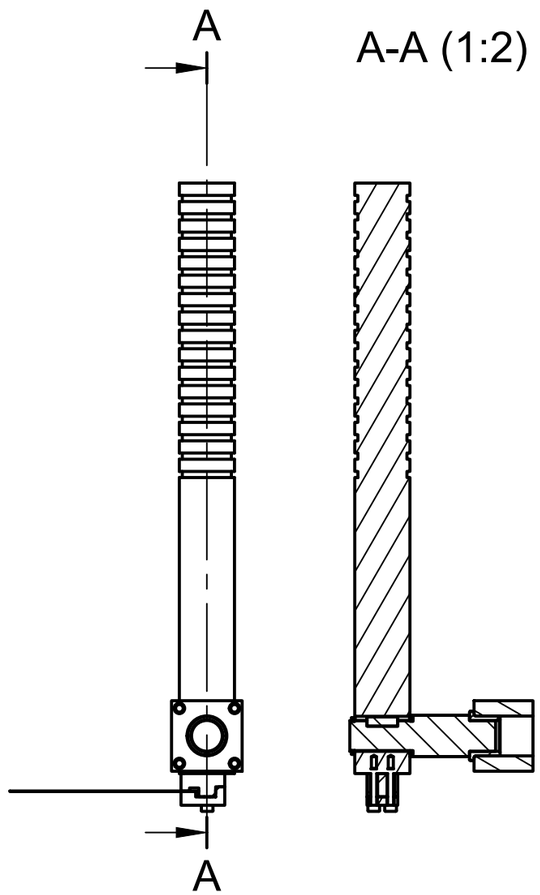




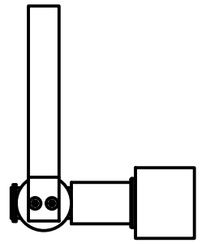
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			Rev. / Date of issue / Sheet: 2/2

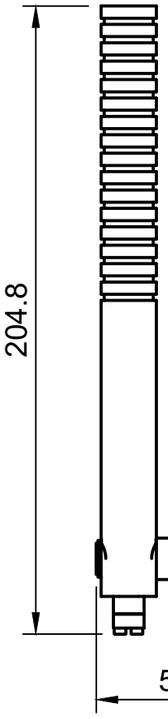
Parts List			
Item	Qty	Part Number	Material
1	1	Blade_B3	Steel
2	1	Lever	Aluminum
3	1	BH_Bielle	Steel
4	1	BH_small_14.6_0.1_pivot	Steel
5	2	DIN EN ISO 4762 - M2 x 12 Steel 4.6 Plain	Steel 4.6, Plain
6	1	Arbre_levier	Steel
7	2	Circlip	Steel
8	1	Key	Steel
9	1	62351700	Bronze
10	1	Bearing support	Steel

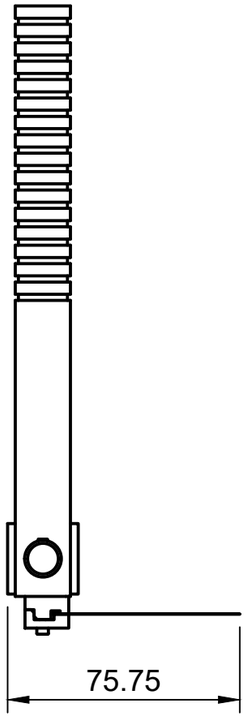


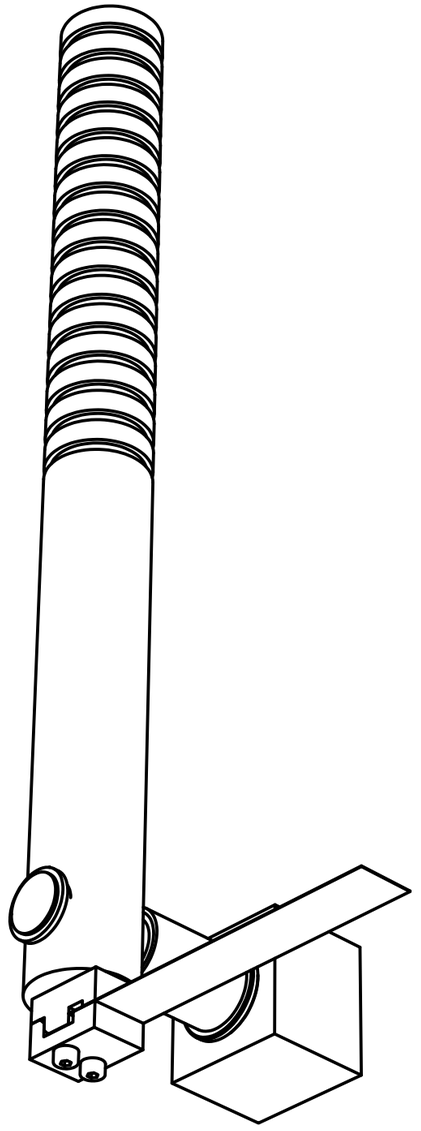


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		Title Levier_asb	DWG No.
		Rev.	Date of issue / Sheet 1/2

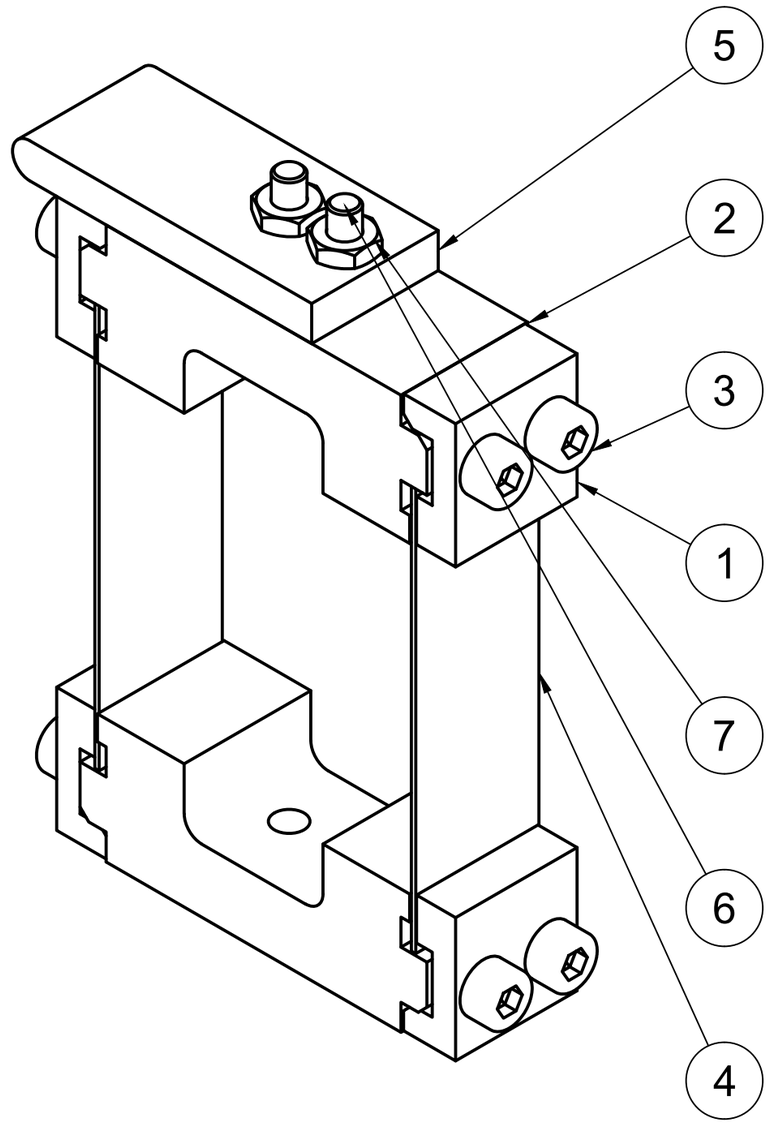








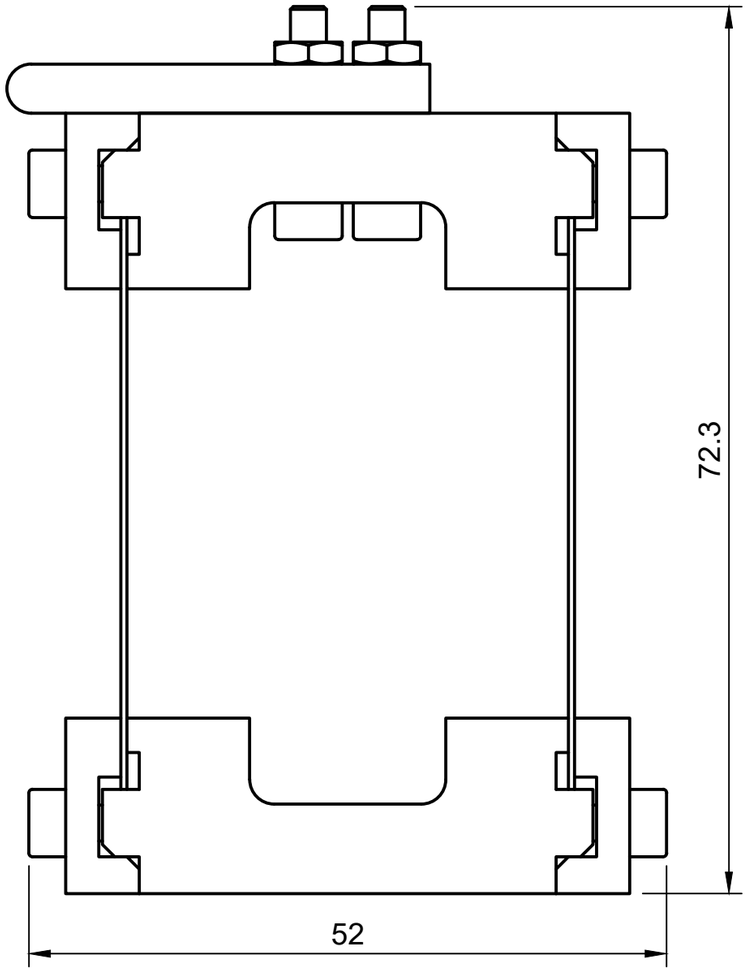
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			Rev. / Date of issue / Sheet 2/2

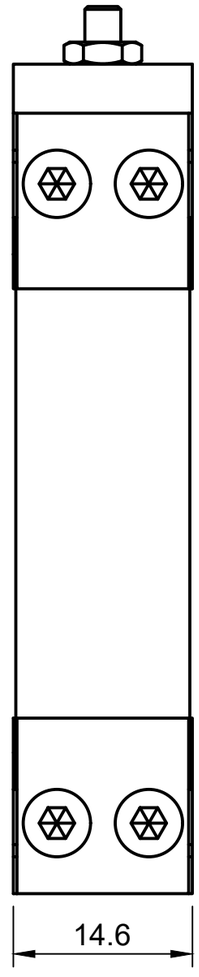


Parts List			
Item	Qty	Part Number	Material
1	4	BH_small_14.6_0.1	Steel
2	2	BH_14.6	Steel
3	8	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
4	2	Blade_13.72	Steel
5	1	Pusher_T2	Steel
6	2	DIN EN ISO 4762 - M3 x 16 Steel 4.6 Plain	Steel 4.6, Plain
7	2	DIN 439-2 - M3 x 0.5 Steel 6 Plain	Steel 6, Plain

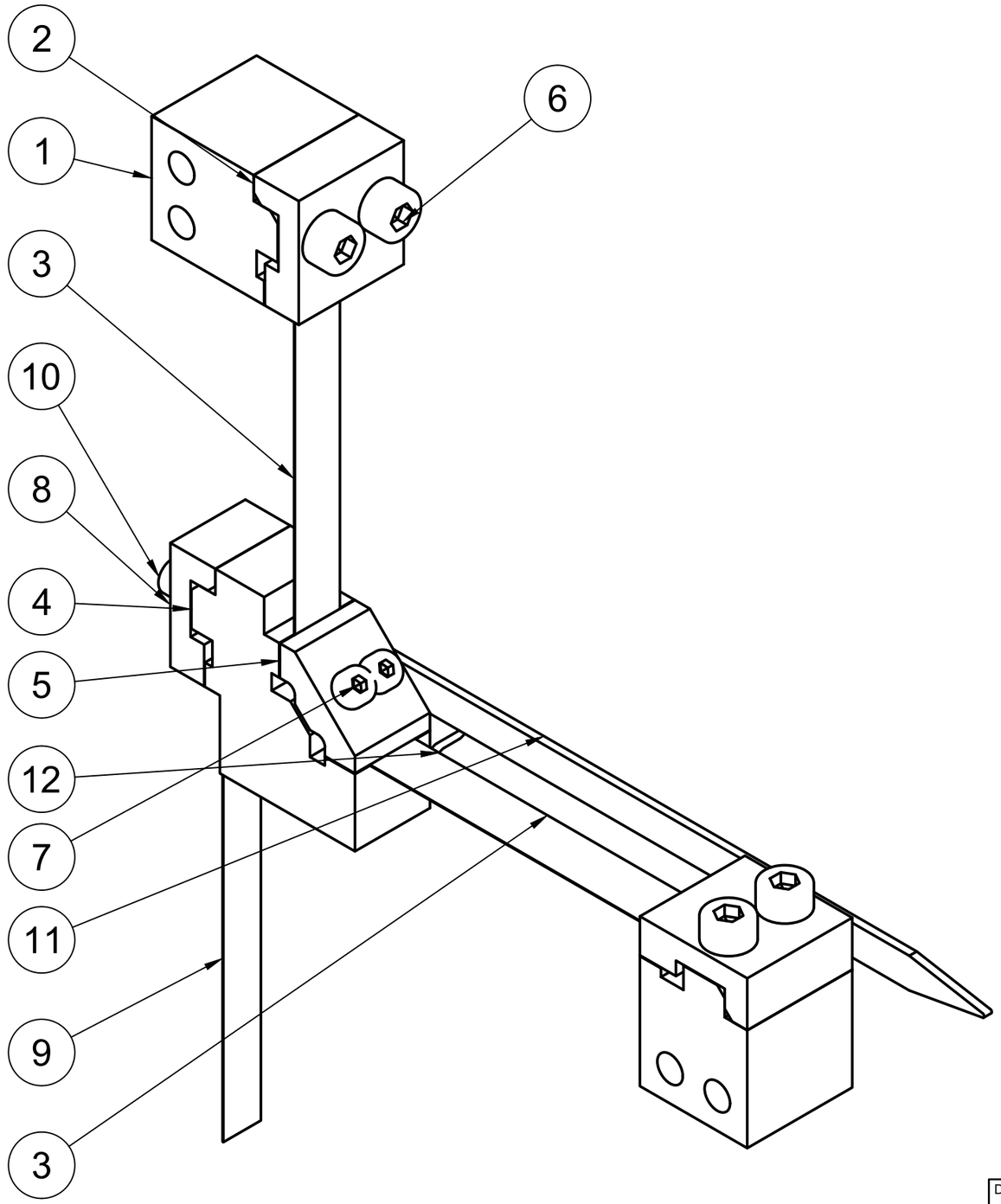
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			Rev. / Date of issue / Sheet 1/2







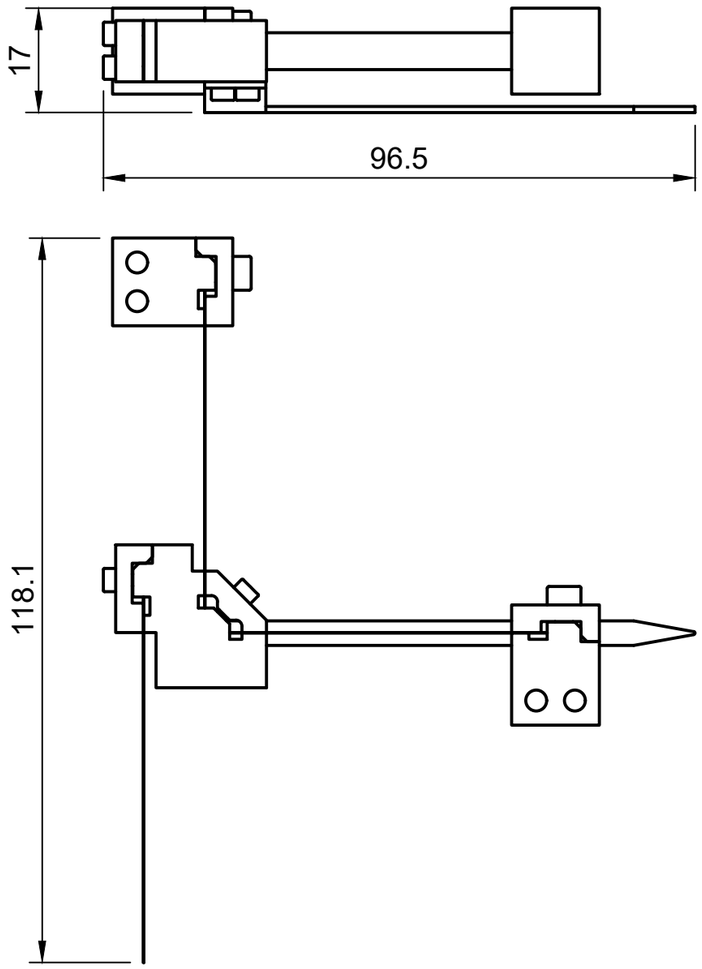
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			Rev. / Date of issue / Sheet 2/2



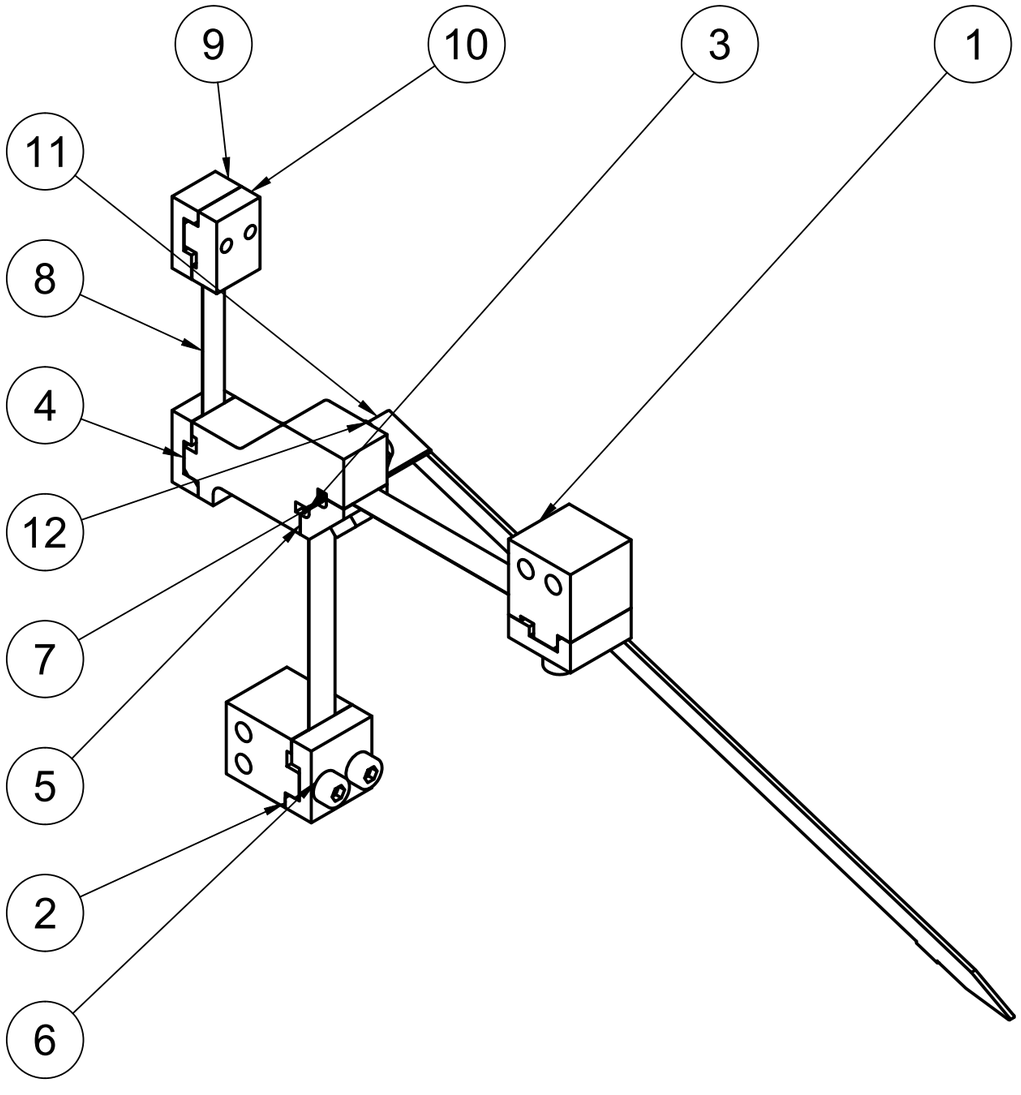
Parts List			
Item	Qty	Part Number	Material
1	2	BH_pivot_14.6	Steel
2	2	BH_small_14.6_0.1	Steel
3	2	Blade_P	Steel
4	1	Hand_80	Steel
5	1	BH_angled_14.6	Steel
6	4	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
7	2	DIN EN ISO 4762 - M2 x 8 Steel 4.6 Plain	Steel 4.6, Plain
8	1	BH_small_14.6_0.1 _pivot	Steel
9	1	Blade_B1	Aluminum
10	2	DIN EN ISO 4762 - M2 x 10 Steel 4.6 Plain	Steel 4.6, Plain
11	1	Aiguille_P1	Steel
12	2	DIN EN ISO 4762 - M2 x 6 Steel 4.6 Plain	Steel 4.6, Plain

Dept.	Technical reference	Created by Dimitri Fahmy 14/05/2026	Approved by
EPFL		Document type	Document status
		Title Hand_asb_P1	DWG No.
			Rev. / Date of issue / Sheet 1/2

Scale 1:1

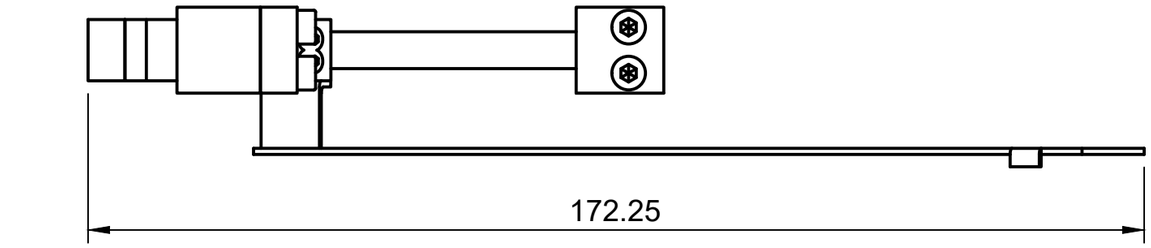


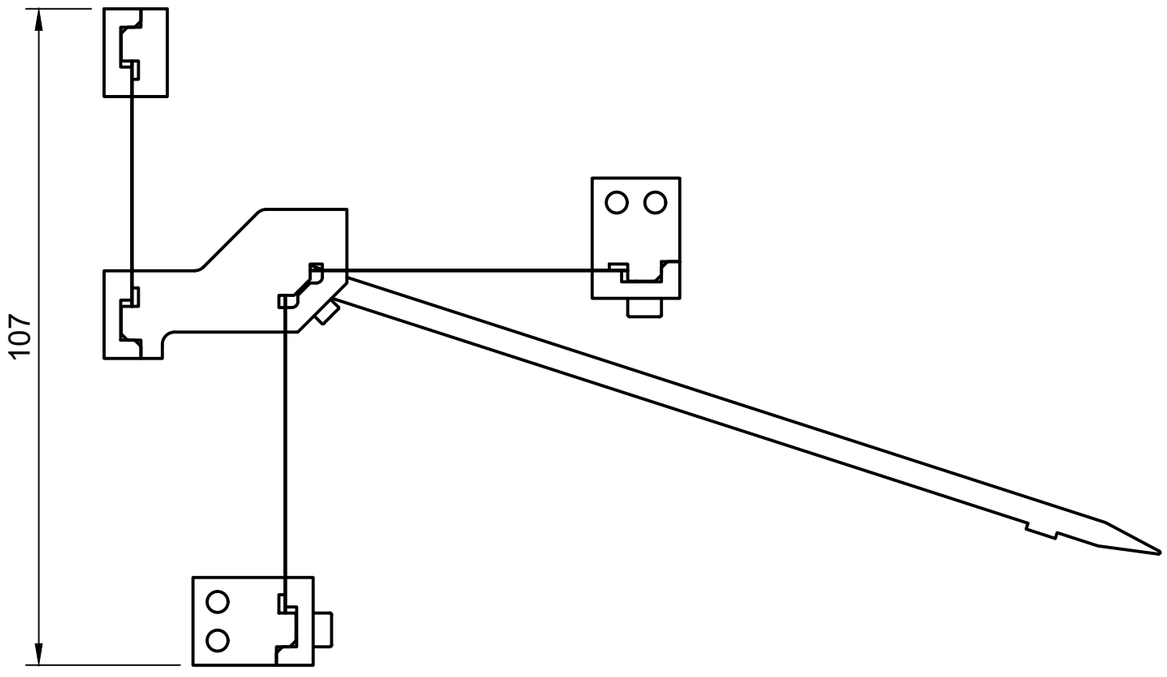
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			Rev. / Date of issue / Sheet 2/2

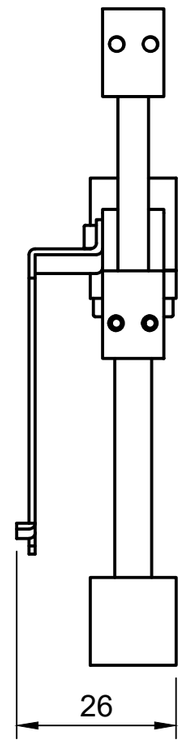


Parts List			
Item	Qty	Part Number	Material
1	2	BH_pivot_14.6	Steel
2	2	BH_small_14.6_0.1	Steel
3	2	Blade_P	Steel
4	1	Pivot_P2	Steel
5	1	BH_angled_14.6	Steel
6	4	DIN EN ISO 4762 - M3 x 10 Steel 4.6 Plain	Steel 4.6, Plain
7	2	DIN EN ISO 4762 - M2 x 8 Steel 4.6 Plain	Steel 4.6, Plain
8	1	Blade_B2	Steel
9	2	BH_small_14.6_0.1 _pivot	Steel
10	1	BH_Bielle	Steel
11	1	Aiguille_P2	Steel
12	2	DIN EN ISO 4762 - M2 x 6 Steel 4.6 Plain	Steel 4.6, Plain

Dept.	Technical reference	Created by Dimitri Fahmy 15/05/2026	Approved by
		Document type	Document status
		Title Hand_asb_P2	DWG No.
			Rev. / Date of issue / Sheet 1/2







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		Document type	Document status
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			Rev. / Date of issue / Sheet: 2/2